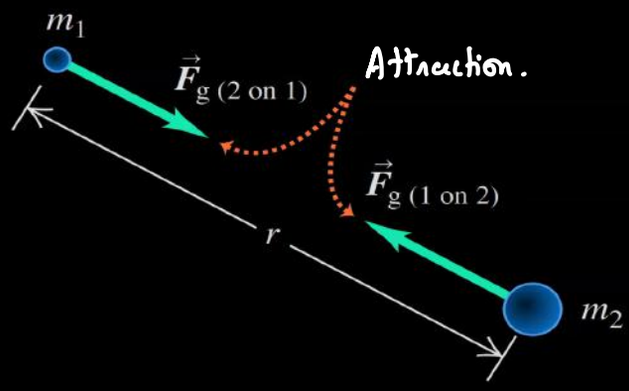
NEWTON'S UNIVERSAL LAW OF GRAVITATION:

Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of masses of the particle and inversely proportional to square of distance between them.



Mathematically:

$$F = \frac{G m_1 m_2}{r^2}$$

Vectorially we can also write:

$$\vec{F}_{21} = \frac{G m_1 m_2}{r_{21}^2} \hat{r}_{12}$$ Where $\vec{F}_{21}$ is force on particle 2 because of particle 1 and $\vec{r}_{21}$ is position of particle 2 wrt 1.

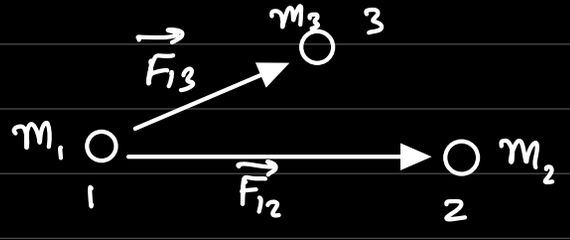
This equation can also be written as follows:

$$\vec{F}_{21} = -\frac{G m_1 m_2}{r_{21}^3} \vec{r}_{21}$$

$G = 6.67 \times 10^{-11} \, N m^2/kg^2$

SUPERPOSITION PRINCIPLE: Force of interaction between any two bodies or particles is independent of presence or absence of any other body in the universe.

Thus $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$



FINDING GRAVITATIONAL FORCE BY INTEGRATION: For finding forces between extended objects, we realise that they are made of several particles and by superposition principle, the net force of interaction can be calculated by integrating the pairwise forces.

Or simply, $F = \int dF$.

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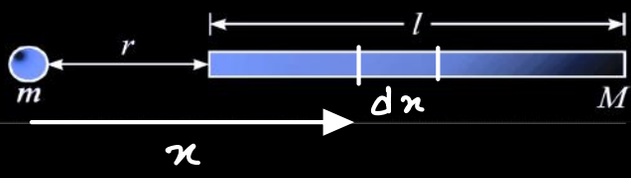
Example:

Find the force of interaction between a point mass m and a rod of mass M.

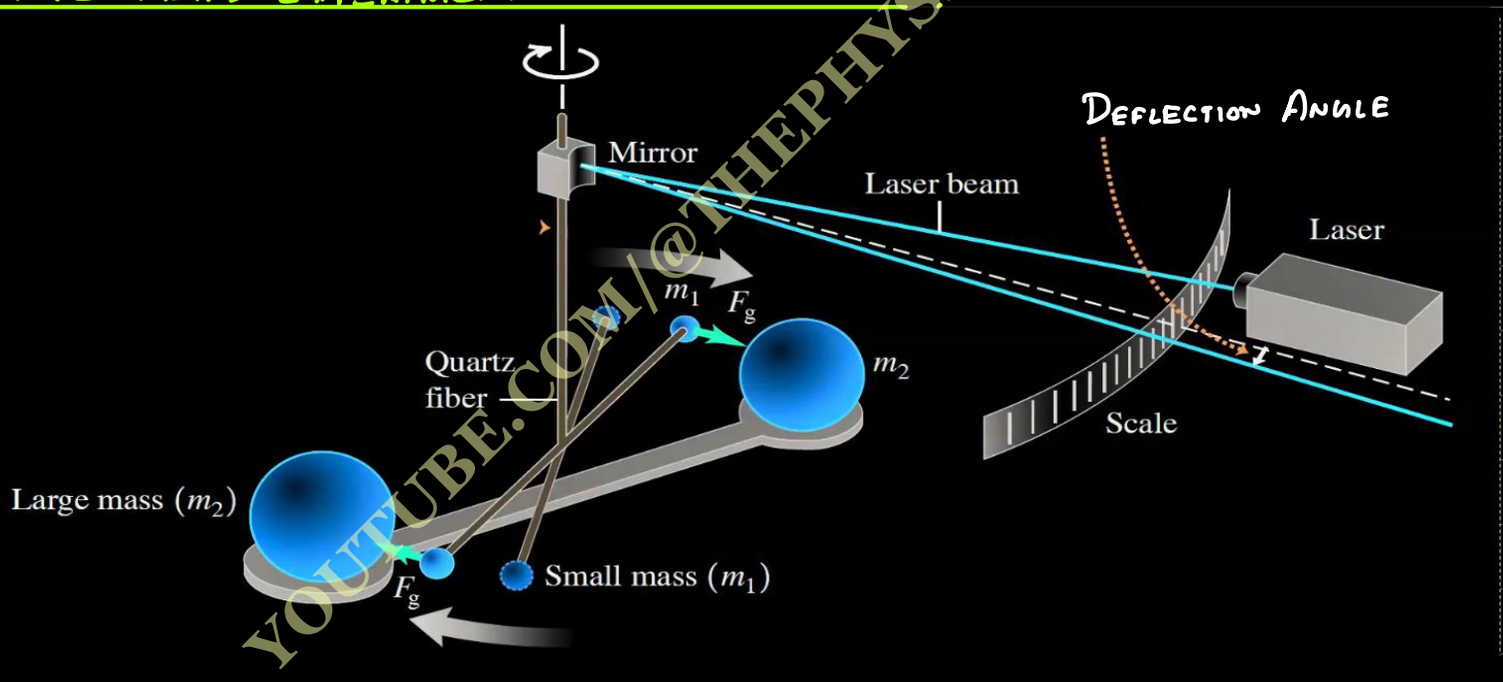
Solution: Consider an element of length dx of the rod at a distance x from point mass m. Now elemental force:



$$dF = \frac{Gm\left(\frac{M}{l}dx\right)}{x^2}$$

$$\Rightarrow F = \frac{GMm}{l}\int_{x}^{x+l}\frac{1}{x^2}dx \Rightarrow F = \frac{GMm}{l}\left[\frac{1}{x} - \frac{1}{x+l}\right] \text{ Ans.}$$

Cavendish's Experiment For Finding G



The Cavendish's set up for finding G is shown above. Essentially it is based on finding the angular deflection of a dumbell like structure suspended from a Quartz fiber due to Gravitational torque generated by massive spheres. To increase the sensitivity in measuring the angular deflection, a laser beam is used, whose from a mirror mounted on Quartz fiber is calibrated on a scale.

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Gravitational Field (Local Acceleration Due to Gravity):

Gravitational force experienced per unit mass kept at any point of space is called Gravitational field at that point. Mathematically if $\vec{F}$ is the gravitational force experienced by a point mass m then Gravitational field at that point is given as:

$$\vec{g} = \frac{\vec{F}}{m} \Rightarrow \boxed{\vec{F} = m\vec{g}}$$

It can be readily seen that it has units of acceleration and also represents the acceleration of point mass m, if it were released free at that point of space. ∴ Gravitational field can be said to be just another name for acceleration due to gravity.

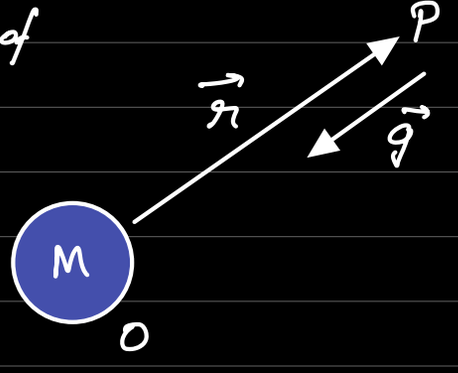
Thus to find Gravitational field at any point, we can imagine a unit mass kept at that point and simply find Gravitational force on it.

Gravitational Potential: Negation of work done by Gravitational forces per unit mass in bringing a particle from a reference point to a given point is called Gravitational potential at that point [This also represents change in Gravitational potential energy of the system per unit mass being transported from reference point to that point]. Typically the reference point is taken at infinity although it may be taken anywhere depending on convenience for solving problems.

Field Due To A Point Mass:

It can be readily seen from our understanding of previous section that field due to a point mass at a general position vector $\vec{r}$ is given as:

$$\vec{g} = -\frac{GM}{r^2}\hat{r} = -\frac{GM\vec{r}}{r^3}$$



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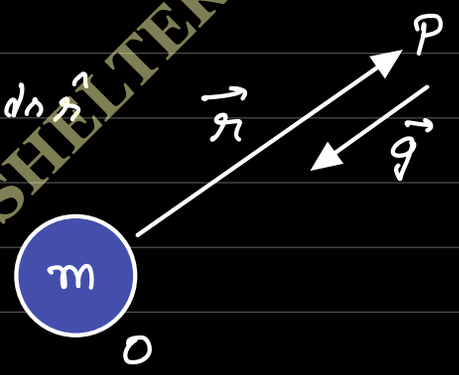
Potential Due To A Point Mass:

Recall the potential at any point in space of point is negation of work done by gravitational force in bringing a unit mass from a reference configuration to the given configuration, thus:

$$V = -\int_{\infty}^{\vec{r}} \vec{g}\cdot d\vec{r}$$ if reference configuration is taken at infinity.

Alternatively we can also write, $V = \int_{\vec{r}}^{\infty} \vec{g}\cdot d\vec{r}$ (∵ Limits Swapped).



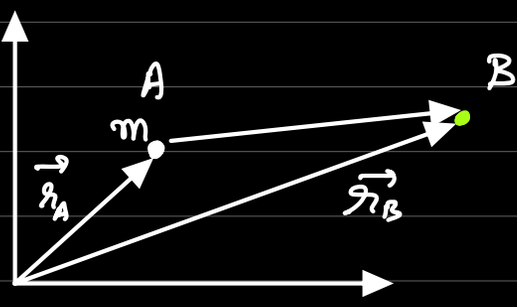
Now for point mass: $\vec{g} = -\frac{Gm}{r^2}\hat{r}$, $d\vec{r} = dr\,\hat{r}$

Thus $V = -\int_{\infty}^{r} \frac{Gm}{r^2}\,dr = \left[\frac{GM}{r}\right]_{\infty}^{r}$

$$\Rightarrow \boxed{V = -\frac{Gm}{r}}$$

Change in Potential Energy using concept of Potential:

We wish to find change in gravitational potential energy as we move a particle of mass m from a point A to point B with other masses in universe being fixed. Now:



$$dU = -m\vec{g}\cdot d\vec{r}$$

Thus our required change in potential energy is given as:

$$\Delta U = -\int_{\vec{r}_A}^{\vec{r}_B} m\vec{g}\cdot d\vec{r}$$

Now since gravitational field is a conservative field, exact path should not matter, so we might as well take a route from A to B through, infinity.

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Thus $\Delta U = -\int_{\vec{r}_A}^{\infty} m\vec{g}\cdot d\vec{r} - \int_{\infty}^{\vec{r}_B} m\vec{g}\cdot d\vec{r} = m\left[-V_A + V_B\right]$

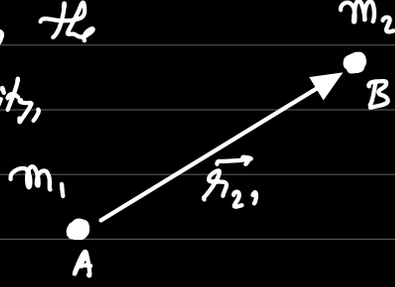
Thus $\Delta U = m(V_B - V_A)$. This illustration shows the utility of finding the potential function. Once we have potential function for a region of space we can immediately find change in potential energy (or work done by external agent without change in KE simply as:

$$\boxed{\Delta U = \Delta W_{ext} = m\Delta V}$$

GRAVITATIONAL INTERACTION ENERGY: For a system of particles, gravitational interaction energy is defined as negation of work done by gravitational forces in assembling the system bringing each mass from infinity. (This is also the change in gravitational PE in the process.

GRAVITATIONAL INTERACTION ENERGY OF A SYSTEM OF TWO PARTICLES:

Consider a system of two particles as shown in the figure. Imagine both particle initially at infinity, now let us bring particle A to the required position. In this process, the gravitational force does no work since there is no force on it. Now when we bring particle B to its final required position, work is done due to field of particle A. In this process change in PE is:



$$U = \Delta U = m_2\Delta V = m_2\left[-\frac{Gm_1}{r_{21}} - 0\right]$$

$$\Rightarrow \boxed{U = -\frac{Gm_1m_2}{r_{21}}}$$

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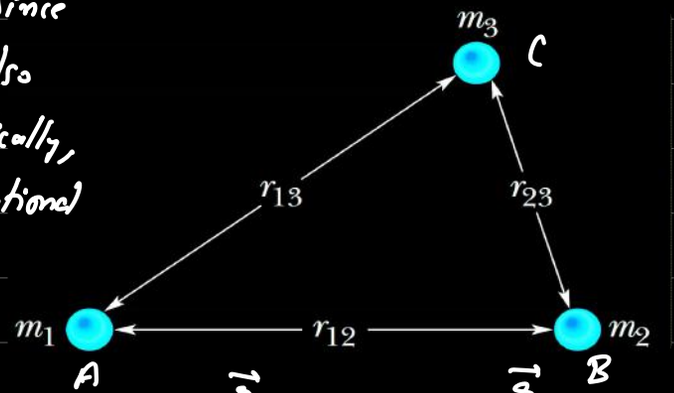
Interaction Energy Of Three Or More Particles:

Note that potential is a scalar. Now since dot product is distributive ∴ potential also follows superposition principle. Mathematically, if $\vec{g}_1$ and $\vec{g}_2$ are individual gravitational fields due to particles 1 and 2, then net potential due to both at any point of space,



$$V = -\int_{\infty}^{\vec{r}} (\vec{g}_1 + \vec{g}_2)\cdot d\vec{r} = -\int_{\infty}^{\vec{r}} \vec{g}_1 \cdot d\vec{r} - \int_{\infty}^{\vec{r}} \vec{g}_2 \cdot d\vec{r}$$

$\Rightarrow V_1 + V_2$. (Same argument can be extended to several particles as well).

Now when a system of three particles is to be assembled as shown in figure above, we throw all the three of them to infinity and bring them one by one to their desired place. Just like in previous case, bringing first particle doesn't change any PE because no field acts on it. When we bring the second particle (say B) it comes in field of A. ∴ In this step change in PE:

$$U_{12} = -\frac{G m_1 m_2}{r_{12}} \quad (1),$$

Now when particles C is brought, it experiences force due to A as well as B. Net potential at location of C due to A and B is:

$$V_3 = \left(-\frac{G m_1}{r_{13}} - \frac{G m_2}{r_{23}}\right)$$

Thus change in PE while bringing m_3, $U_{23} + U_{13} = m_3 V_3$

$$\Rightarrow U_{23} + U_{13} = -\frac{G m_1 m_3}{r_{13}} - \frac{G m_2 m_3}{r_{13}} \quad (2)$$

Thus total potential energy of the system is, obtained by adding equations (1) and (2):

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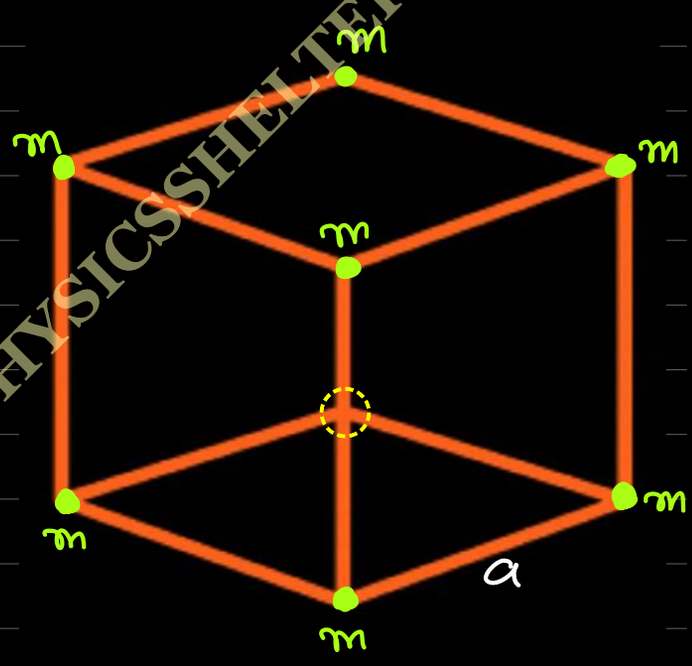
$$U = U_{12} + U_{13} + U_{23} = -\left[\frac{Gm_1m_2}{r_{12}} + \frac{Gm_2m_3}{r_{23}} + \frac{Gm_3m_1}{r_{13}}\right]$$

The same argument can be extended to a system of n particles also. For a system of n particles:

$$\boxed{U = -\sum \frac{Gm_im_j}{r_{ij}}}$$

Here each pairwise interaction must be counted only once

EXAMPLE: Seven out of eight vertices of a cube have a point mass placed on them as shown. Find the gravitational PE of the system. Also find work required to place another point mass on the vacant vertex bringing m from infinity.



Solution: Finally all vertices have masses, we shall count the interaction by categories:

(I) Along edges, $(m, m, a) = 12$

(II) Along face diagonals $(m, m, a\sqrt{2}) = 12$

(III) Along body diagonals $(m, m, a\sqrt{3}) = 4$

Thus final potential energy: $U_f = -\frac{Gm^2}{a}\left[12 + \frac{12}{\sqrt{2}} + \frac{4}{\sqrt{3}}\right]$

Now when one of the masses was missing, the interactions will be:

(I) Along edges, $(m, m, a) = 9$

(II) Along face diagonals $(m, m, a\sqrt{2}) = 9$

(III) Along body diagonals $(m, m, a\sqrt{3}) = 3$

Thus $U_i = -\frac{Gm^2}{a}\left[9 + \frac{9}{\sqrt{2}} + \frac{3}{\sqrt{3}}\right]$

Thus $\Delta U = U_f - U_i = -\frac{Gm^2}{a}\left[3 + \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}}\right]$ Ans.

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(Alternately we could also count the new interactions introduced by placing the point mass on the vacant vertex) thus:

$$\Delta U = \frac{-Gm^2}{a}\left[3 + \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}}\right] \quad \because \text{ it is involved in 3 edges,}$$

3 faces and 1 Body diagonal.

Relation Between Gravitational Fields & Potential :

Recall the differential form of potential equation, $dV = -\vec{g}\cdot d\vec{r}$

Let $\vec{g} = g_x\hat{i} + g_y\hat{j} + g_z\hat{k}$, Now suppose if I move || to x axis keeping y & z coordinates fixed by some elemental displacement given as $d\vec{r} = dx\,\hat{i}$, then: $dV = -\vec{g}\cdot d\vec{r}$

$\Rightarrow dV = -g_x dx \Rightarrow g_x = -\frac{dV}{dx}$, but since V is a 3D function here, actually by $\frac{dV}{dx}$ we mean here partial derivation

$\because$ We kept y & z fixed. Thus:

$$\boxed{g_x = -\partial V/\partial x}$$ Similarly:

$$\boxed{g_y = -\partial V/\partial y}$$ and

$$\boxed{g_z = -\partial V/\partial z}$$

Thus we can also write $\vec{g} = -\left[\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right]$

or in compact form $\boxed{\vec{g} = -\vec{\nabla}V}$ where $\vec{\nabla}$ is an operator defined as

$$\boxed{\vec{\nabla} = \left[i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right]}$$

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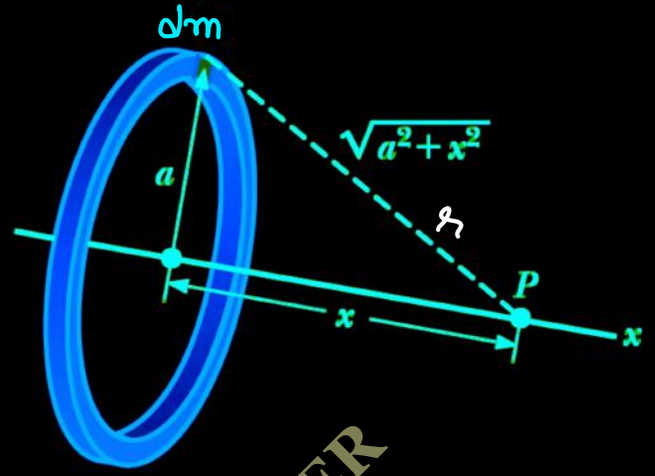
Potential And Field On Axis Of A Ring:

Consider a ring of mass m and radius a, and a point P at a distance x from its centre on the axis. Due to a mass dm, the potential at point P is given as:



$$dV = \frac{-G\,dm}{\sqrt{a^2+x^2}}$$

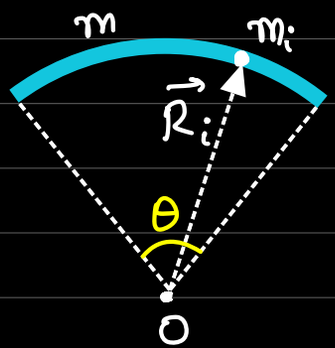
$$\Rightarrow V = \int dV = \frac{-GM}{\sqrt{a^2+x^2}} \Rightarrow \boxed{V = \frac{-GM}{\sqrt{a^2+x^2}}}$$

Now $g_x = -\frac{dV}{dx} = \frac{-GMx}{(a^2+x^2)^{3/2}}$, thus:

$$\boxed{g_x = \frac{-GMx}{(a^2+x^2)^{3/2}}}$$ ← This has maximum magnitude at $x = a/\sqrt{2}$.

Potential And Field At Centre Of Curvature Of An Arc:

Consider an arc of mass m and radius R, let m_i be mass of its i^{th} particle located at a position vector $\vec{R}_i$ wrt its centre of curvature O. Now due to this particle:



$$\vec{g}_i = \frac{Gm_i}{R^2}\hat{R}_i = \frac{Gm_i\vec{R}_i}{R^3}$$

$$\Rightarrow \vec{g} = \frac{G}{R^3}\sum m_i\vec{R}_i \times \frac{m}{m} = \frac{Gm}{R^3}\vec{r}_{CM}$$ where $\vec{r}_{CM}$ is the location of CM of the arc. Now recall from chapter on CM, $\vec{r}_{CM}$ for arc is given by $r_{CM} = \frac{R\sin(\theta/2)}{(\theta/2)}$

Thus for arc $\boxed{\vec{g} = \frac{GM}{R^2}\frac{\sin(\theta/2)}{(\theta/2)}\hat{n}}$ ($\hat{n}$ is along bisector).

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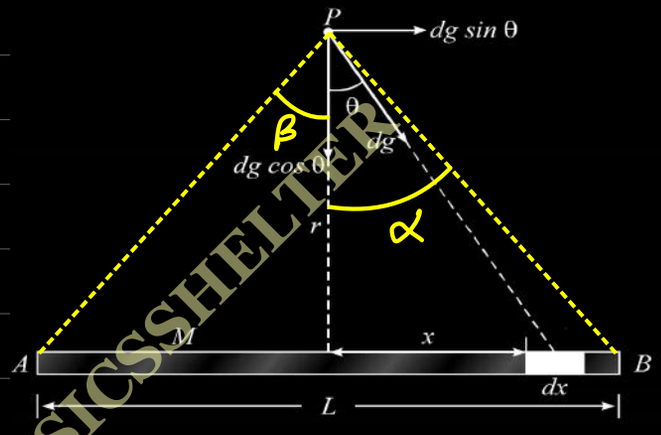
Now for potential recognise that each particle of arc is equidistant from centre of curvature. Potential due to i^{th} particle, $V_i = \frac{-Gm_i}{R}$

$\Rightarrow V = \sum V_i = -Gm/R$ Thus $\boxed{V = \frac{-GM}{R}}$

Field Due To A Line Segment (Off Axis Point) :

Consider a rod of mass M and a point P at a distance r from it subtending angles α and β at extremities as shown. Now consider an element of length dx at a distance of x from foot of $\perp$ from P.



Now :

$$dg = -G\left(\frac{M}{L}\right)dx \cdot \frac{1}{(r\sec\theta)^2}$$

Also $x = r\tan\theta \Rightarrow dx = r\sec^2\theta\, d\theta$, thus,

$$dg = -\frac{GM}{Lr^2\sec^2\theta} \times r\sec^2\theta\, d\theta \Rightarrow dg = -\frac{GM}{Lr}\, d\theta .$$

Now field is a vector quantity so we must integrate its components separately, field component $\perp$ to rod, $dg_\perp = dg\cos\theta$ and $dg_{||} = dg\sin\theta$

Thus:

$$\int dg_\perp = -\frac{GM}{Lr}\int_{-\beta}^{\alpha} \cos\theta\, d\theta = -\frac{GM}{Lr}\left[\sin\alpha + \sin\beta\right]$$

Here $-$ sign signifies the attractin field (direction towards rod).

Thus :

$$\boxed{g_\perp = \frac{GM}{Lr}\left[\sin\alpha + \sin\beta\right]}$$

$$\text{Similarly } \int dg_{||} = \int_{-\beta}^{\alpha} \frac{GM}{Lr}\sin\theta\, d\theta = \frac{GM}{Lr}\left[\cos\beta - \cos\alpha\right]$$

Thus, $\boxed{g_{||} = \frac{GM}{Lr}\left[\cos\beta - \cos\alpha\right]}$

→ Remember here α is along +ve x axis and β along −ve x axis.

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Field Due To Infinite Wire:

For infinite wire (both sides ∞),

$\alpha = \beta = 90^\circ \Rightarrow g_\perp = \frac{GM}{Lr}[\sin 90^\circ + \sin 90^\circ] = \frac{2GM}{Lr}$ and $g_\parallel = 0$

Thus $\vec{g} = -\frac{2GM}{Lr}\hat{r} = -\frac{2G\lambda}{r}\hat{r}$ where $\lambda = M/L$ (Linear Density).

Potential Due To Infinite Wire:

Recall, $\vec{g} = -\nabla V$, thus $g_r = -\frac{dV}{dr}$

now applying this to infinite wire, $-\frac{2G\lambda}{r} = -\frac{dV}{dr}$

$\Rightarrow V = 2G\lambda \ln(r) + C$ where C can be arbitrary constant.

Here if we take $V=0$ @ $r=\infty$ then C is not defined ($\because \ln(\infty)$ is not defined) so for convenience we choose potential to be zero at some distance $r = r_0$ from the rod, thus:

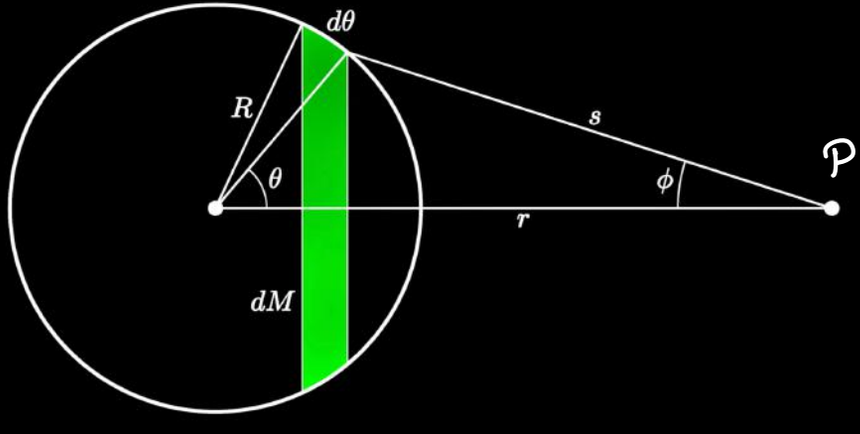
$0 = 2G\lambda \ln(r_0) + C \Rightarrow C = -2G\lambda \ln(r_0)$,

using this our potential function becomes:

$$\boxed{V = 2G\lambda \ln\left(\frac{r}{r_0}\right)}$$

Potential Due To A Thin Spherical Shell:

If you have studied electrostatics before, you would immediately recognise the parallels between almost all mathematical results of electrostatics with gravitation. (No worries if you haven't, it is not a pre requisite, though doing gravitation after electrostatics is much easier because by and large the theory is a subset of electrostatics theory). In electrostatics we derived field and potential of shell using Gauss's law, we could use the same approach here also, however we will see an alternate approach here.



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Consider an element in shape of cap of a sphere as shown in the figure above. For the element, $dM = \frac{M}{4\pi R^2} \times 2\pi R \sin\theta \times R\,d\theta = \frac{M}{2}\sin\theta\,d\theta$

Now for a point located at r, potential is given as:

$$V = \frac{-G\,dM}{s}$$

Now s and θ both vary with element we shall transform everything to variable s.

Now using cosine law, $s^2 = R^2 + r^2 - 2Rr\cos\theta \Rightarrow 2s\,ds = 2Rr\sin\theta\,d\theta$

$\Rightarrow dM = \frac{M}{2} \times \frac{s\,ds}{Rr}$, thus, $dV = -G \times \frac{M}{2} \times \frac{s\,ds}{Rr} \cdot \frac{1}{s} = \frac{-GM\,ds}{2Rr}$.

Now for an outside point, s varies $r-R$ to $r+R$, thus,

$$V_{out} = \frac{-GM}{2Rr}\int_{r-R}^{r+R} ds = \frac{-GM}{r} \Rightarrow \boxed{V_{out} = \frac{-GM}{r}}$$

For inside point, s varies from $R-r$ to $R+r$ as θ varies from 0 to π. Thus:

$$V_{in} = \frac{-GM}{2Rr}\int_{R-r}^{R+r} ds = \frac{-GM}{R}$$

thus $\boxed{V_{in} = \frac{-GM}{R}}$

→ Note that potential at outside point is like that of a point mass

→ Potential at inside point is constant.

FIELD DUE TO A THIN UNIFORM SPHERICAL SHELL:

Recall the relation, $g = \frac{-dV}{dr}$, using this we can readily derive the expressions for gravitational field outside and inside:

$$g_{in} = \frac{-dV_{in}}{dr} = 0, \quad g_{out} = \frac{-d}{dr}\left[\frac{-GM}{r}\right] = \frac{-GM}{r^2}$$

thus: $\boxed{g_{in} = 0}$ $\boxed{\vec{g}_{out} = \frac{-GM}{r^2}\hat{r}}$

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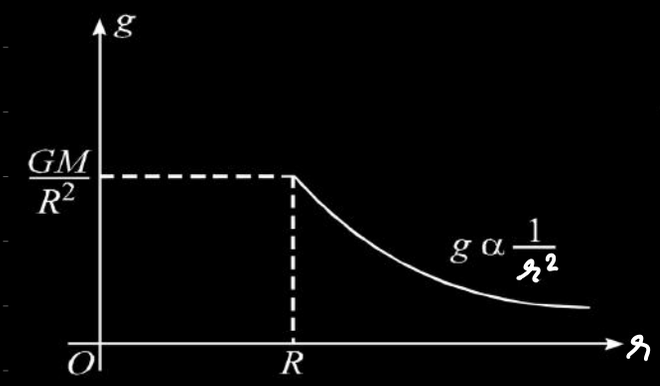
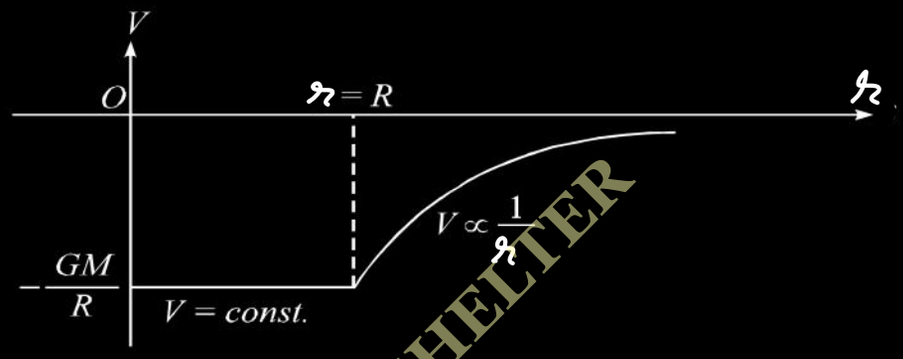
Shell Theorems:

Shell theorem I: For an inside point a uniform shell produces no gravitational field.

Shell theorem II: For an outside point a shell behaves just like a point mass at its own centre.

Field & Potential Plots For Shell :

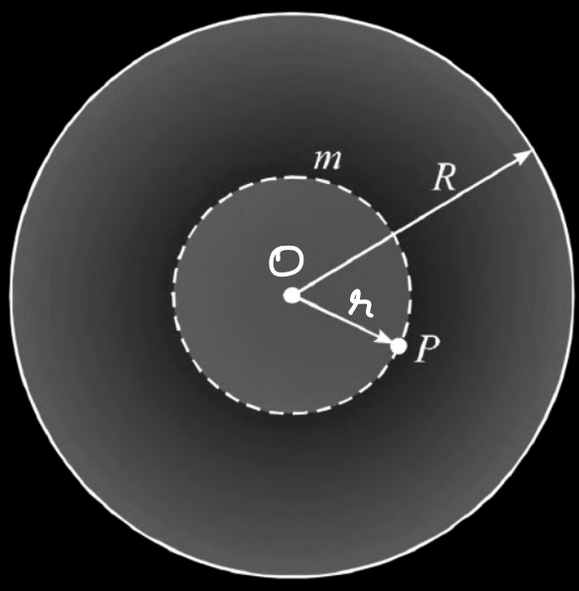




Field Due To A Uniform Solid Sphere:

A solid sphere can be visualised to be made of several concentric thin shells, thus by shell theorem II, for outside points, the entire sphere can be assumed to be concentrated at its own centre.

Thus for $r > R$, $\boxed{g = -\frac{GM}{r^2}}$



Now let us consider an inside point P ($r < R$), Here by shell theorems, the portion between r & R contributes nothing to the field and the portion between O & r behaves like a point mass.

Thus:

For $r < R$, $g = -G\left(\frac{M}{\frac{4}{3}\pi R^3}\right) \times \left(\frac{4}{3}\pi r^3\right) \cdot \frac{1}{r^2} = -\frac{GMr}{R^3}$

thus: $\boxed{\vec{g} = -\frac{GMr}{R^3}\hat{r}}$ also, $\rho = \frac{M}{\frac{4}{3}\pi R^3} \Rightarrow M = \frac{4}{3}\pi R^3 \rho$

using this, $\boxed{\vec{g} = -\frac{4\pi G}{3}\rho\vec{r}}$

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Potential Due To A Uniform Solid Sphere:

For outside points, by shell theorem, the potential is just like that of a point mass, i.e. for $r > R$, $\boxed{V = -\frac{GM}{r}}$

For surface, $r = R$, $\boxed{V = -\frac{GM}{R}}$

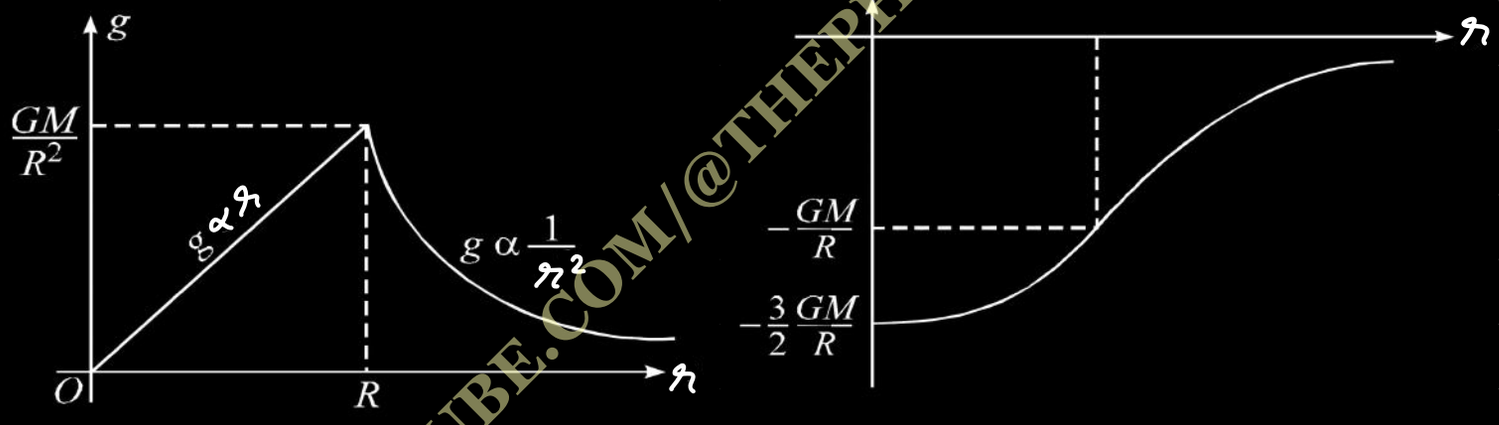
Now for inside points we can use the fact that $g = -dV/dr$.

Thus:

$$-dV/dr = -GMr/R^3 \Rightarrow \int_{-\frac{GM}{R}}^{V} dV = \int_{R}^{r} \frac{GMr}{R^3} dr \quad (\because @\, r = R,\ V = -GM/R)$$

$$\Rightarrow V + \frac{GM}{R} = \frac{GM}{2R^3}\left[r^2 - R^2\right] \Rightarrow \boxed{V = -\frac{GM}{R}\left[1.5 - 0.5\frac{r^2}{R^2}\right]}$$

Field & Potential Plots For Solid Sphere:

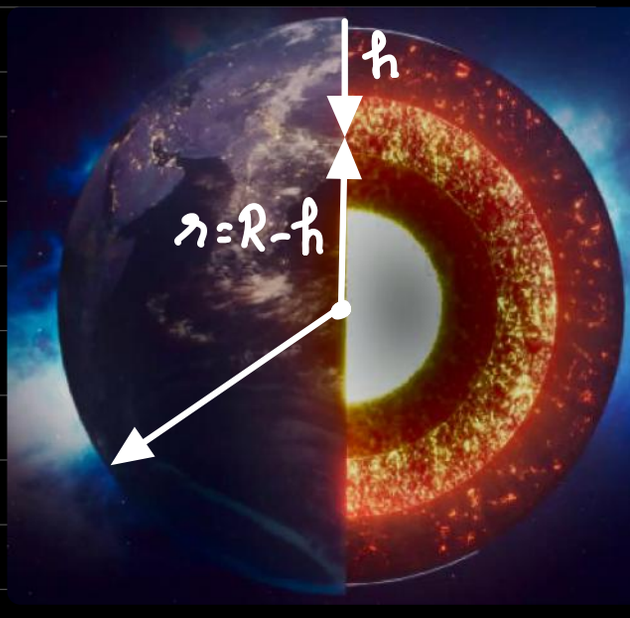


Variation Of Earth's Gravity With Depth:

For most purposes, gravitational field of earth can be approximated by that of a uniform solid sphere. Thus for a point at depth h, $\vec{g} = -\frac{GMr}{R^3}\hat{r}$

thus $\vec{g}' = -\frac{GM(R-h)}{R^3}\hat{r}$

or $\vec{g}' = -\frac{GM}{R^2}\left[1 - \frac{h}{R}\right]\hat{r}$



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Now at surface, $g = \frac{GM}{R^2}$,

thus $\boxed{g' = g[1 - h/R]}$

Variation Of Earth's Field With Height :

As derived earlier, $g' = \frac{GM}{r^2}$ for $r > R$,

thus for earth, $g' = \frac{GM}{(R+h)^2}$

$$\Rightarrow g' = \frac{GM}{R^2}\left(1+\frac{h}{R}\right)^{-2} = g\left(1+\frac{h}{R}\right)^{-2}$$

Now if $h << R$, we can use binomial approximation, $\boxed{g' = g\left(1 - \frac{2h}{R}\right)}$

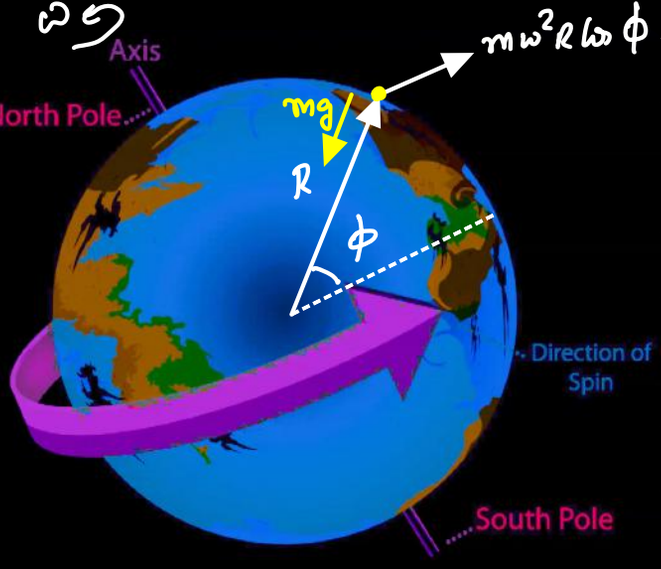
Thus for small distances, the acceleration due to gravity declines twice as fast with height as with depth (assuming uniform density, which is actually not true).

Variation Of Acceleration Due To Gravity With Latitude :

Due to rotation of earth any particle on surface of earth also experiences a centrifugal force apart from the gravitational attraction, the effective gravity is resultant of the two forces thus, in radial direction,

$$g' = g - \frac{m\omega^2 R\cos^2\phi}{m}$$

$$\Rightarrow \boxed{g' = g - \omega^2 R\cos^2\phi}$$



→ Thus effective gravity is maximum at poles and minimum at equator.

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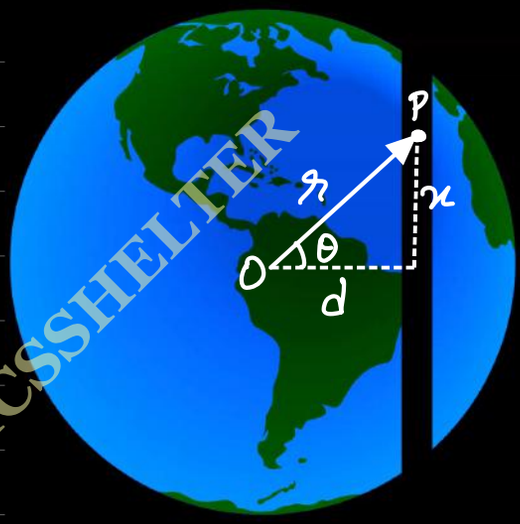
→ Value of ω is very small, $\omega = \frac{2\pi}{24\times60\times60}$ rad s^{-1} so ω^2 is smaller still, thus for earth this effect is not very significant.

→ Apart from radial component, centrifugal force also has a component in tangential direction, however it doesn't cause any significant change in magnitude of total effective force.

SHM IN EARTH TUNNEL:

Imagine a tunnel dug through earth in to which a particle is released. Let the particle be at a distance x from the point in the tunnel nearest to earth's centre. The force on the particle along the tunnel is



$$F_{11} = \frac{GM}{R^3} r \times m \times \sin\theta = ma$$

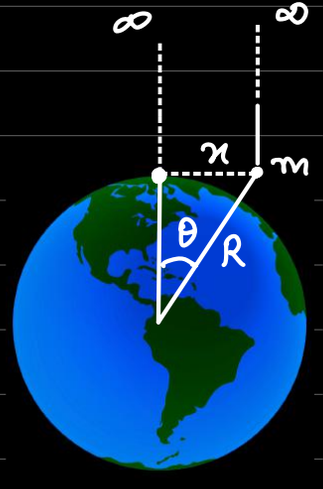
$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{GMx}{R^3}$$, this is equation of SHM, thus the particle executes SHM with time period:

$$T = 2\pi\sqrt{\frac{R^3}{GM}} = 2\pi\sqrt{\frac{R}{g}} \quad \left(\because \frac{GM}{R^2} = g\right).$$

→ After substituting the values for earth, the time comes out to be 84.6 minutes, thus from anywhere to anywhere, travel time along the tunnel from surface to surface point would be 42.3 minutes (T is independent of d).

TIME PERIOD OF INFINITE LENGTH PENDULUM:

If the pendulum string has infinite length, for small displacements, centre of curvature is at ∞ thus motion can be assumed to be along straight line (tangent of earth), now restoring force:



$$F = mg\sin\theta = mg\frac{x}{R} = -ma \Rightarrow \boxed{T = 2\pi\sqrt{\frac{R}{g}}}$$

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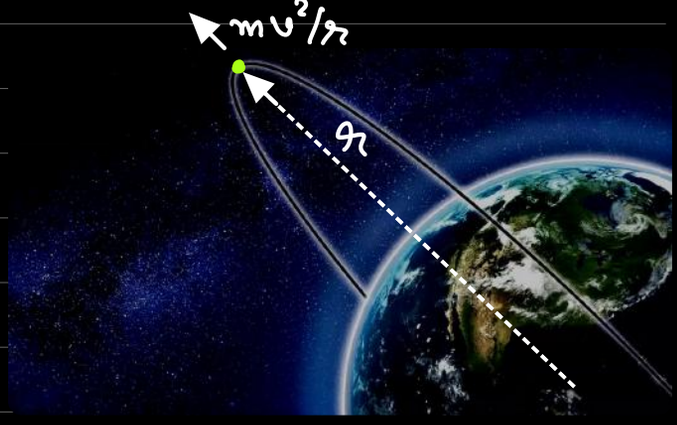
Thus such a pendulum also has a time period of 84.6 minutes. (Effectively this will also be time period of oscillation in a frictionless pipe fixed on surface of earth).



ORBITAL VELOCITY The velocity that must be given to a satellite so that it may execute a circular orbit of certain radius around a planet is called orbital velocity. Let mass of planet be M, now balancing centrifugal and gravitational forces:

$$\frac{mv_o^2}{r} = \frac{GMm}{r^2} \Rightarrow v_o = \sqrt{\frac{GM}{r}}, \quad v_o = \text{Orbital velocity.}$$

In case of earth, $g = \frac{GM}{R^2} \Rightarrow v_o = \sqrt{\frac{gR^2}{r}} = v_s\sqrt{\frac{R}{r}}$ where

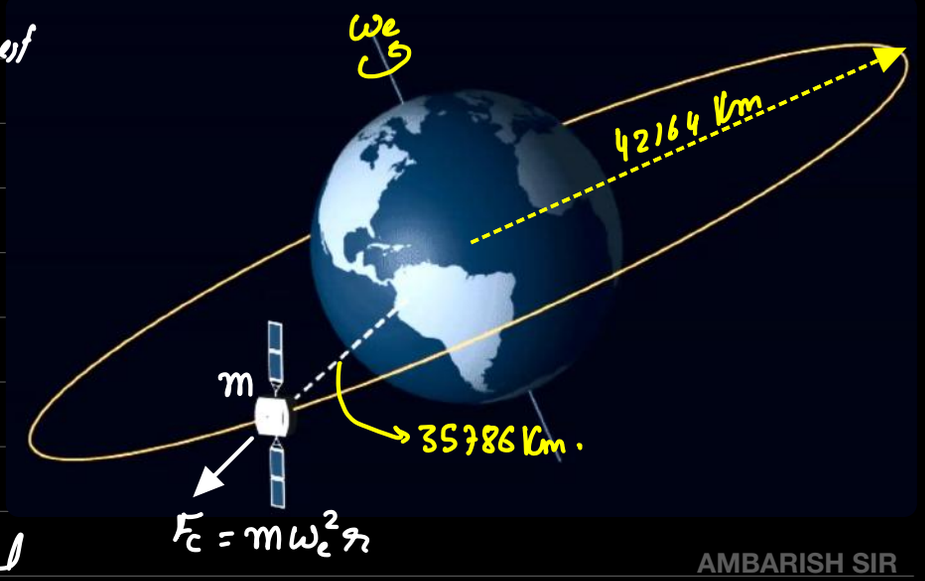
v_s is the orbital velocity for a satellite near the surface of the earth i.e $v_s = \sqrt{gR} \approx 7.9\ km\,s^{-1}$.

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{r}} = v_s\sqrt{\frac{R}{r}}$$

$$v_s = \sqrt{gR} = \sqrt{\frac{GM}{R}} \approx 7.9\ km/sec.$$

GEOSTATIONARY SATELLITE:

A satellite which is always at rest with respect to the rotating frame of earth is called geostationary satellite. For a geostationary satellite, the angular velocity must match the angular velocity of earth. A geostationary satellite can only be present in equatorial



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plane of earth, because gravitational force must pass through centre of earth. Let ω_e be the angular velocity of earth, now balancing centrifugal force by gravitational force:

$$m\omega_e^2 r = \frac{GMm}{r^2} \Rightarrow r = \left[\frac{GM}{\omega_e^2}\right]^{1/3}$$

thus $$\boxed{r_G = \left[\frac{GM}{\omega_e^2}\right]^{1/3}} \quad (1)$$

Now for earth, $M \approx 6\times10^{24}$ Kg, $\omega_e = 2\pi/24\times3600$,

we get $r_G = 42164$ Km from centre of earth, now radius of earth is 6400 Km, to get the altitude from surface, we need to subtract radius of earth. Thus:

$$\boxed{h_G \approx 36000} \text{ Km from surface}$$

→ From equation (1) above, we can say that $\boxed{T^2 = \frac{4\pi^2 r^3}{GM}}$

where T is orbital period of any satellite in orbit of radius r. this can be readily used to compare time periods of two satellites in various circular orbits, i.e $T_1^2/T_2^2 = r_1^3/r_2^3$.

→ If a satellite revolves around a planet in the same sense as the rotation of the planet, the motion is said to be prograde.

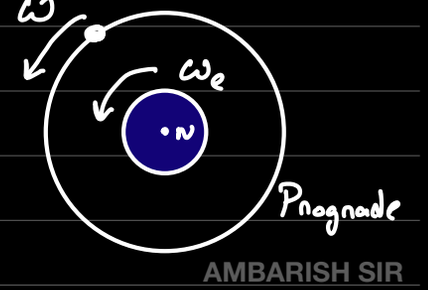
→ If a satellite revolves around a planet in opposite sense to the rotation of planet, we call the motion retrograde.

→ Geosats are used for communication, weather forecasting, oceanography, tracking

EXAMPLE: A satellite revolves around earth in equatorial plane with angular velocity ω as seen from space frame, If angular velocity of earth's rotation is ω_e find the time interval between the times when satellite is overhead above a certain city on equator if the satellite is (a) Prograde (b) Retrograde.

Solution: From rotating frame of earth, a prograde satellite in equatorial plane will appear to be revolving with angular velocity $\omega_{eff} = \omega - \omega_e$

Thus for a prograde satellite, overhead times



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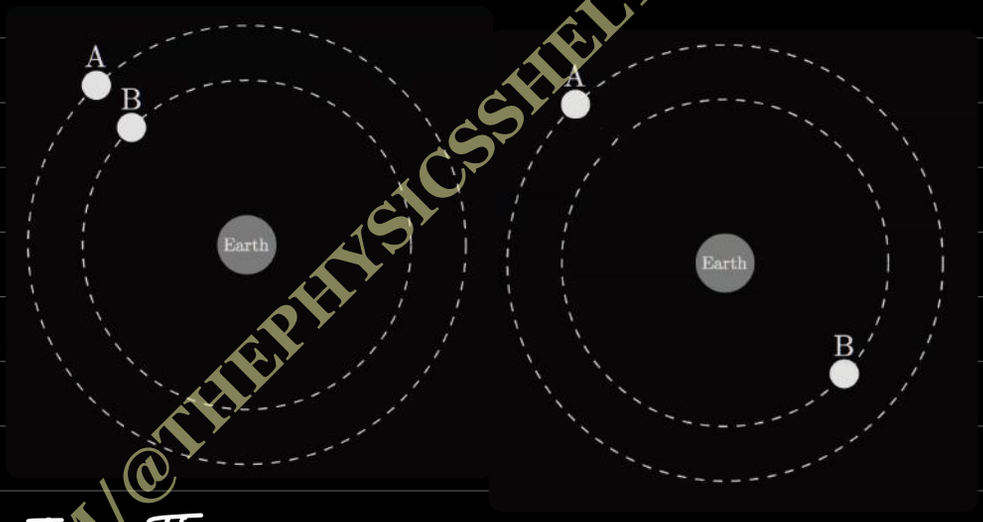
over any city on equator, $T_{eff} = \frac{2\pi}{\omega_{eff}} = \frac{2\pi}{|\omega - \omega_e|}$ Ans

Similarly for a retrograde satellite, $\omega_{eff} = \omega + \omega_e$

$\Rightarrow$ Time interval between overhead times : $T_{eff} = \frac{2\pi}{\omega + \omega_e}$ Ans.

EXAMPLE: Two satellites revolve around earth with angular speeds ω_A and ω_B in space frame, both of them are moving in equatorial plane, find the time period between times when satellites are collinear with earth if: (a) Both are prograde (b) One is prograde and other is retrograde satellite.

Solutions: Silly mistake alert! There are two possible collinear configurations:



Thus relative to one satellite the other satellite has to rotate only 180° for moving from one collinear position to another collinear position. Thus when both are prograde satellites, required time period : $T_c = \frac{\pi}{|\omega_A - \omega_B|}$

If one is prograde and other one retrograde, then $T = \frac{\pi}{\omega_A + \omega_B}$

→ Prograde and retrograde motions can also be defined for satellites in non equatorial planes. If $\vec{\omega}$ of satellite make acute angle with $\vec{\omega}_e$ if call it prograde, if it is an obtuse angle we call it retrograde satellite.

→ Prograde Satellites are launched near the equator to take advantage of the natural speed due to earth's rotation.

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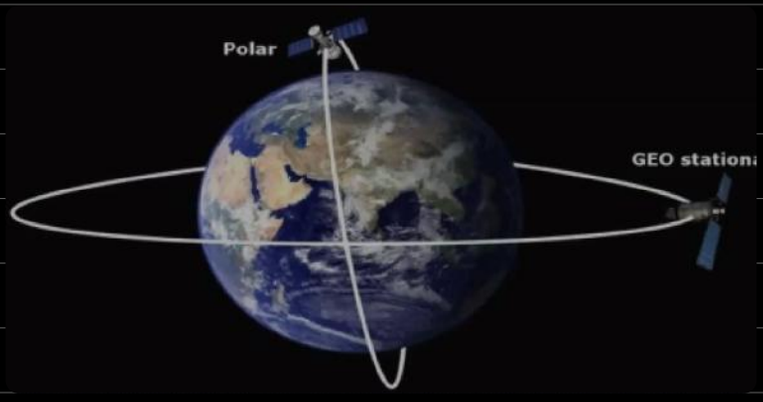
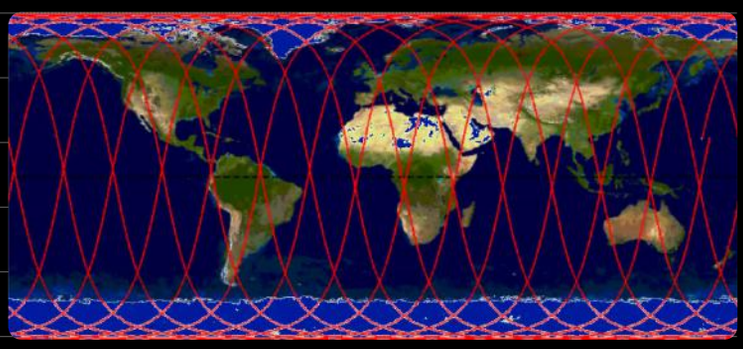
Polar Satellites : A satellite whose orbit passes from above poles of earth is called a polar satellite thus for polar satellites, $\vec{\omega} \perp \vec{\omega}_e$. Such a satellite will generally pass from above various points on earth, thus unless the time period of polar satellite is exact multiple or fraction of earth's rotation time period, it will pass overhead above every point of earth. Since earth rotates from west to east, polar satellite would appear to drift westwards from its ground track longitude.

→ Polar satellites are typically used for mapping earth, reconnaisance (spying) and also for weather monitoring and predictions. They are also used for remote sensing (Classifying objects on earth without coming in physical contact).



Ground Tracking.



Ground Track Map

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Example: Pathfinder Mc03:

Suppose the Saturn rings consist of football size particles orbiting in circular orbits. If the maximum speed of these particles is 1.05 times of the minimum speed, find ratio of radial width to the inner radius of the ring.

(a) 0.01 (b) 0.05
(c) 0.10 (d) 0.25

Solution: $v_0 = \sqrt{\frac{GM}{r}} \Rightarrow \ln v_0 = \ln \sqrt{GM} - \frac{1}{2} \ln r$

$\Rightarrow \frac{1}{v_0} dv_0 = -\frac{1}{2r} dr$ (1), Now let inner radius of ring be r

and outer radius be $r+dr$, thus ① $\Rightarrow \frac{1-1.05}{1} = -\frac{1}{2}\frac{dr}{r}$

$\Rightarrow \frac{dr}{r} = 0.1$ dms. Thus ⓒ is correct (note that at smaller r v is larger).

EXAMPLE : PATHFINDER MCQ 4 :

Two planets A and B are orbiting around their sun in circular orbits of radii r and $4r$. In a quarter year of planet B, how many revolutions will the planet A make?

(a) 1 (b) 2
(c) 4 (d) 8

Solution: Recall that $T^2 = \frac{4\pi^2}{GM} r^3$, now number of revolutions in a given time period is inversely proportional to the time period, thus

$$\frac{N_A^2}{N_B^2} = \frac{r_B^3}{r_A^3} \quad \text{or} \quad N_A = N_B\left(\frac{r_B}{r_A}\right)^{3/2} = \frac{1}{4}(4)^{3/2} = 2$$

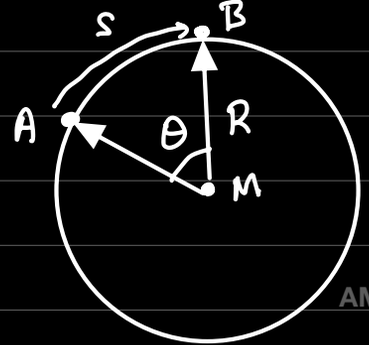
thus A makes 2 revolutions during quarter revolution of planet B.
correct option is (b)

EXAMPLE: PATHFINDER : BUILD-6 :

Two space stations A and B are orbiting the earth with speed v in the same orbit of radius R. Station B is a distance s ahead of station A. Two astronauts from station A have to join the station B. For this purpose, they leave station A in a spaceship and move radially to a slightly lower orbit and stay in that orbit and when the station B comes closest to them, they immediately move radially to join the station B. If time spend in the lower orbit is Δt, find separation between the orbits.

Solution: $v = \sqrt{\frac{GM}{R}} \Rightarrow \omega_A = \omega_B = \omega$

$\omega = \sqrt{\frac{GM}{R^3}}$ ① also $\theta = \frac{s}{R}$ ②



Now let lower orbit be located at $R-\epsilon$

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Thus ω in lower orbit: $\omega' = \sqrt{\frac{GM}{(R-\epsilon)^3}}$. Now time required to cover the angle $\theta = \Delta t = \frac{\theta}{(\omega' - \omega)}$, thus:

$$\Delta t = \frac{S/R}{\sqrt{GM}\left[(R-\epsilon)^{-3/2} - R^{-3/2}\right]} = \frac{S}{vR^{3/2}\left[(R-\epsilon)^{-3/2} - R^{-3/2}\right]} \quad (\because \sqrt{GM} = v\sqrt{R})$$

$$\Rightarrow \Delta t = \frac{S}{v}\left[\frac{1}{(1-\epsilon/R)^{-3/2} - 1}\right] = \frac{S}{v} \cdot \frac{2R}{3\epsilon} \quad (\because \epsilon << R)$$

$$\Rightarrow \epsilon = \frac{2SR}{3v\Delta t} \text{ Ans.}$$

Total Energy In Circular Orbit:

Recall that potential due to a solid sphere at an outside point is given as $V = -\frac{GM}{r}$, thus potential energy of a satellite in a circular orbit:

$$U = mV \Rightarrow \boxed{U = -\frac{GMm}{r}}$$

Also $v = \sqrt{\frac{GM}{r}} \Rightarrow KE = \frac{1}{2}mv^2 \Rightarrow \boxed{KE = \frac{GMm}{2r}}$

Now total Energy $= KE + U = -\frac{GMm}{2r}$

Thus $\boxed{E = -\frac{GMm}{2r} = -KE = \frac{U}{2}}$

Angular Momentum In Circular Orbit:

$v = \sqrt{\frac{GM}{r}} \Rightarrow L$ of satellite about centre of planet:

$L = mvr = m\sqrt{\frac{GM}{r}} \times r = m\sqrt{GMr}$, thus $\boxed{L = m\sqrt{GMr}}$

We can also show that areal velocity of satellite is equal to $L/2m$. (Areal velocity means area swept per unit time).

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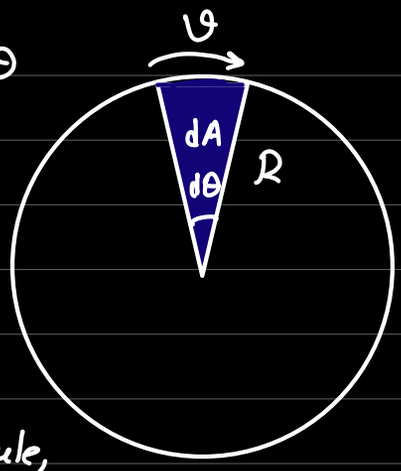
Consider adjacent figure, $dA = \left(\frac{d\theta}{2\pi}\right) \times \pi R^2 = \frac{1}{2} R^2 d\theta$



thus $\frac{dA}{dt} = \frac{1}{2} R^2 \frac{d\theta}{dt}$, $L = mR^2\omega = mR^2 \frac{d\theta}{dt}$

thus $\boxed{\frac{dA}{dt} = \frac{L}{2m}}$ or $\boxed{\frac{d\vec{A}}{dt} = \frac{\vec{L}}{2m}}$

Here area vector is defined by right hand thumb rule, i.e curl the fingers of right hand along sense of rotation, then thumb gives the direction of area vector.

EXAMPLE : IRODOV 1·227 :

An artificial satellite of the Moon revolves in a circular orbit whose radius exceeds the radius of the Moon η times. In the process of motion the satellite experiences a slight resistance due to cosmic dust. Assuming the resistance force to depend on the velocity of the satellite as $F = \alpha v^2$, where α is a constant, find how long the satellite will stay in orbit until it falls onto the Moon's surface. Assume that while spiralling down, the orbit is always nearly circular.

Solution: Method I : WORK ENERGY APPROACH : Power developed by resistance forces, $P = \vec{F} \cdot \vec{v} = -\alpha v^3$. Now by work energy theorem this power must be equal to rate of loss of mechanical energy of the system, thus: $-\alpha v^3 = dE/dt$, also $E = -GMm/2R$, thus we have differential equation:

$$-\alpha\left(\sqrt{\frac{GM}{R}}\right)^3 = \frac{d}{dt}\left[\frac{-GMm}{2R}\right] \Rightarrow \frac{GMm}{2R^2}\frac{dR}{dt} = \frac{-\alpha GM\sqrt{GM}}{R^{3/2}}$$

$$\Rightarrow \frac{m}{2}\int_{\eta R}^{R} R^{-1/2} dR = \int_0^{T_f} -\alpha\sqrt{GM}\, dt \Rightarrow m\sqrt{R}\left[1-\sqrt{\eta}\right] = -\alpha\sqrt{GM}\, T_f$$

$$\Rightarrow T_f = \frac{m\sqrt{R}}{\alpha\sqrt{GM}}\left(\sqrt{\eta}-1\right) \Rightarrow T_f = \frac{m}{\alpha\sqrt{gR}}\left(\sqrt{\eta}-1\right) \text{Ans} \quad \left(\because \frac{GM}{R^2} = g\right)$$

Method II Angular momentum approach:

Resistive force is tangential thus continuously reduces angular momentum of the satellite.

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Thus: $-\alpha v^2 \times r = \frac{d}{dt}\left(m\sqrt{GMr}\right) = m\sqrt{GM}\cdot\frac{1}{2\sqrt{r}}\frac{dr}{dt}$

$\Rightarrow \frac{dr}{dt} = \frac{1}{m\sqrt{GM}} \times 2\sqrt{r} \times -\alpha\left(\frac{GM}{r}\right)\times r = \frac{-2\alpha r^{1/2}}{m}\sqrt{GM}$

$\Rightarrow m\int_R^{\eta R} r^{-1/2}\,dr = -\alpha\sqrt{GM}\int_0^{T_f} dt$, this is same integral as

obtained towards end in previous approach and gives same result.

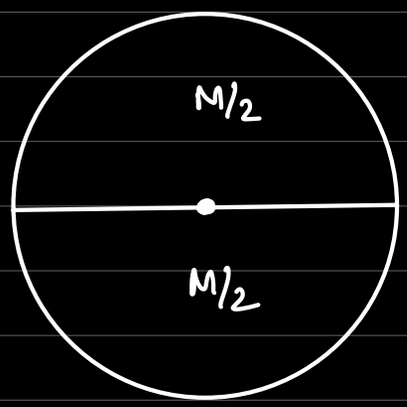
EXAMPLE : IRODOV 1.213:

A particle of mass m was transferred from the centre of the base of a uniform hemisphere of mass M and radius R into infinity.

Find the work required in the process.

Solution. Recall that potential at the centre of a complete sphere is given by: $V = -\frac{3GM}{2R}$, this must be superposition of potentials of two hemispheres each of mass $M/2$. Thus potential at centre due to hemisphere of mass $M/2$:

$V_H = -\frac{3G(M/2)}{2R}$, thus if we have a hemisphere of mass M instead of $M/2$, $M/2$ must be replaced by M. thus $V = -\frac{3GM}{2R}$ at centre of curvature of solid hemisphere.



Now work done by external agent in slow moving: $W_{ext} = \Delta U = m\Delta V$

$= m(V_\infty - V) = m\left[0 - \left(-\frac{3GM}{2R}\right)\right] = \frac{3GMm}{2R}$ Ans.

EXAMPLE : IRODOV 1.205:

A double star is a system of two stars moving around the centre of inertia of the system due to gravitation. Find the distance between the components of the double star, if its total mass equals M and the period of revolution T. (Assume Circular Orbits).

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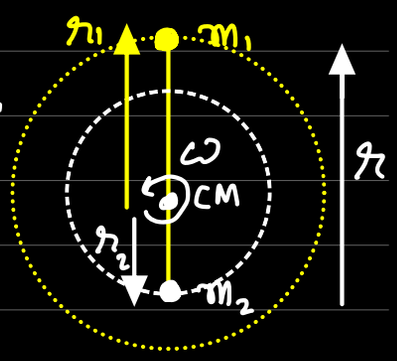
Solution: From CM frame the particles will execute circular orbits about common CM. From CM frame, momentum of each particle is μv_{rel} where μ is $\frac{m_1 m_2}{m_1+m_2}$



and v_{rel} is magnitude of speed of one particle relative to another. Now applying newton's law to particle 1:

$$\frac{G m_1 m_2}{r^2}(-\hat{r}) = \mu \frac{d}{dt}(\vec{v}_{12}) = \mu \omega^2 r (-\hat{r})$$

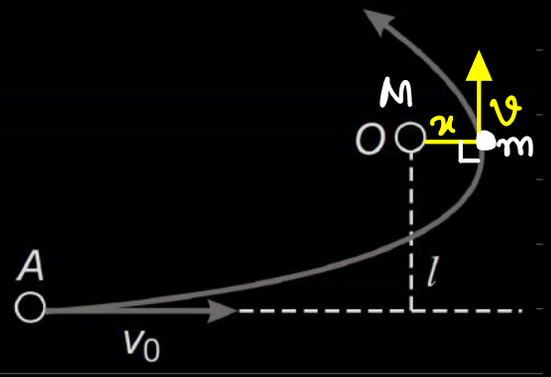
$\because \frac{d}{dt}(\vec{v}_{12})$ is acceleration of particle 1 wrt 2 i.e $\omega^2(r_1+r_2)(-\hat{r})$

Thus $\omega^2 = \frac{G(m_1+m_2)}{r^3} = \frac{GM}{r^3} = \frac{4\pi^2}{T^2}$ $(\because \omega = 2\pi/T)$

$$\Rightarrow r = \left[\frac{GMT^2}{4\pi^2}\right]^{1/3} \text{ Ans.}$$

EXAMPLE : IRODOV 1.210 :

A cosmic body A moves to the Sun with velocity v_0 (when far from the Sun) and aiming parameter l the arm of the vector $\mathbf{v}_0$ relative to the centre of the Sun (Fig. 1.51.). Find the minimum distance by which this body will get to the Sun.



Solution: $\because$ Gravitational force always passes through centre of Sun $\therefore$ torque of gravitational force about sun is zero, thus angular momentum about sun must remain conserved. Thus:

$$m v_0 l = m v x \quad (1)$$ here v is speed at nearest distance and x is the corresponding nearest distance. Now conserving mechanical Energy:

$$\frac{1}{2}mv_0^2 + 0 = \frac{-GMm}{x} + \frac{1}{2}mv^2 \quad (2)$$

Solving (1) & (2):

$$x = \frac{GM}{v_0^2}\left[\sqrt{1+\left(\frac{v_0 l^2}{GM}\right)^2} - 1\right] \text{ Ans.}$$

Here M is mass of the sun.

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Escape Velocity :

$$v_{escape} = 11.2\ km/s$$

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

$$v_{escape} = \sqrt{\frac{2GM}{r}}$$

Minimum velocity with which a projectile must be launched from space so that it may go up to infinite distance from a planet is called escape velocity. Suppose a projectile is launched from a distance r from centre of earth to go all the way to infinity, then in the limiting case, by mechanical energy conservation:

$$-\frac{GMm}{r} + \frac{1}{2}mv_e^2 = 0 \Rightarrow v_e = \sqrt{\frac{2GM}{r}} = \sqrt{2}\,v_o$$

Here v_e is escape velocity and v_o is orbital velocity. For a near earth satellite, $r \approx R = 6400$ Km, for this we get $\boxed{v_e = 11.2 \text{ Km/sec.}}$

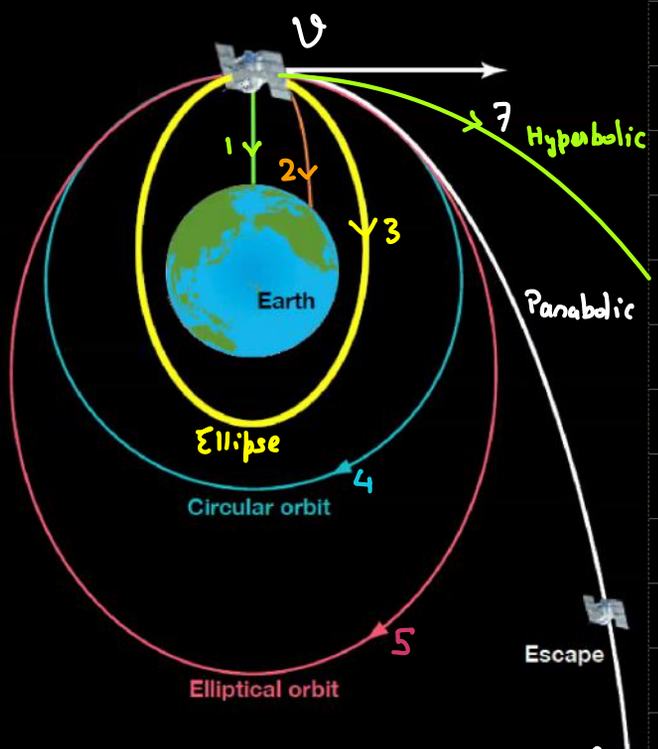
→ Note that escape velocity does not depend on angle of launching, any particle launched above horizon would escape so long as its velocity is more than escape velocity (assuming no drag losses).

→ Exactly at escape velocity, trajectory is parabolic, for velocities greater than escape velocity, the escape trajectory is hyperbolic.

Projectile Trajectories ($v \perp r$) :

Any projectile launched from anywhere under influence of gravity of a planet always forms a conic section (Ellipse, Circle, parabola or a hyperbola). This can be proved using differential equation in polar form ($\vec{F} = m\vec{a}$). Here we state the results for launching $v \perp r$ without proofs: (One of the foci is centre of earth).

1. Dropping from rest: Line Segment path (Line segment is ellipse with foci at end points).
2. $v <$ orbital velocity insufficient to form orbit due to crash : Section of Ellipse



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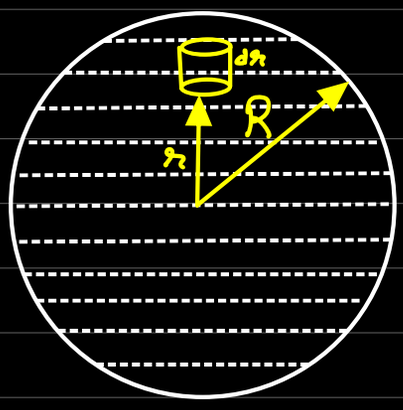
(Launch point being Apogee or farthest point of the orbit).

3. $V < V_0$ sufficient velocity to avoid crash : Elliptical orbit with launch point being Apogee (farthest point of the orbit)
4. $V = V_0$: Circular orbit
5. $V_0 < V < V_e$: Elliptical orbit, with launch point being perigee (nearest point of the orbit).
6. $V = V_e$: Parabolic trajectory, particle escapes to infinity.
7. $V > V_e$: Hyperbolic trajectory, particle escapes to infinity.

EXAMPLE : IRODOV 1.216: GRAVITATIONAL PRESSURE :

A uniform sphere has a mass M and radius R. Find the pressure **p** inside the sphere, caused by gravitational compression, as a function of the distance r from its centre. Evaluate **p** at the centre of the Earth, assuming it to be a uniform sphere.



Solution: Recall from fluid mechanics, $dp = -\rho g\, dy$ where dy is change in height across a fluid element. Now in case of earth we model it like a huge liquid sphere, recall that acceleration due to gravity inside earth varies as $g = \frac{GMr}{R^3}$ Thus:

$$dp = -\rho \frac{GMr}{R^3} dr \quad (- \text{ sign since as we move up, pressure decreases}).$$

$$\Rightarrow \int_{P_0}^{P} dp = -\frac{\rho GM}{R^3} \int_{R}^{r} r\, dr \quad \Rightarrow \quad P = P_0 + \frac{\rho GM}{2R^3}\left[R^2 - r^2\right]$$

Here P_0 is atmospheric pressure. $\rho = \frac{M}{4/3\,\pi R^3}$

$$\Rightarrow \quad P = P_0 + \frac{3}{8\pi}\frac{GM^2}{R^6}\left[R^2 - r^2\right], \text{ If } P_0 = 0,$$

$$\boxed{P = \frac{3}{8\pi R^4} GM^2 \left[\ \ - \frac{r^2}{R^2}\right]} \quad \text{Ans.}$$

→ This is the fluid pressure created in response to Gravitational force so perhaps it will be more appropriate to call it hydrostatic pressure.

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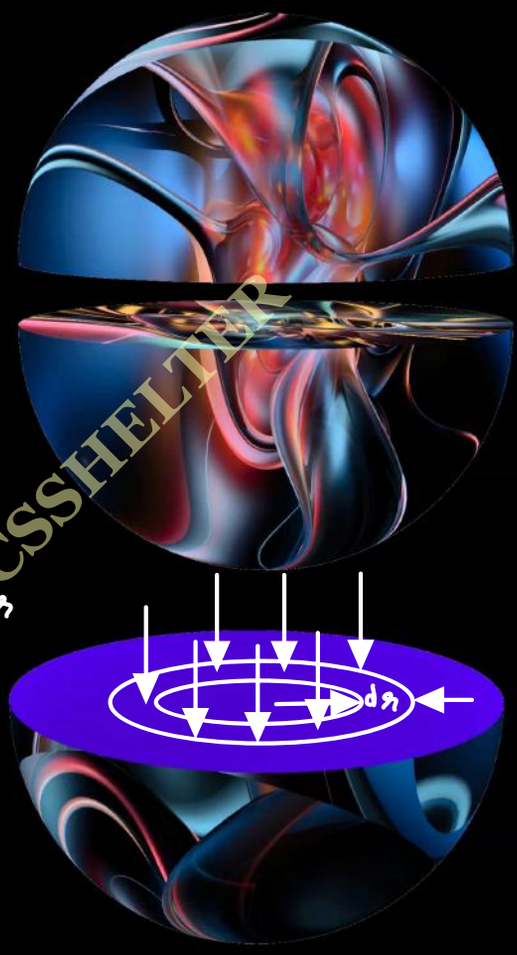
EXAMPLE: GRAVITATIONAL FORCE BETWEEN SOLID HEMISPHERES:

Find the gravitational force of attraction between two halves of a solid sphere of mass M and Radius R.

Solution: Method I: Integrating Gravitational pressure:

We know that gravitational attraction between two hemispheres must be exactly balanced by the hydrostatic pressure pushing force in the equatorial plane, thus we can find the Gravitational pull by Integrating hydrostatic pressure in the equatorial plane.

Now consider an annular region of thickness dr at a distance r from the centre.



The force of hydrostatic repulsion:

$$dF = p \times 2\pi r\,dr = \frac{3GM^2}{8\pi R^4}\left(1-\frac{r^2}{R^2}\right) \times 2\pi r\,dr$$

$$\Rightarrow F = \frac{3GM^2}{4R^4}\int_0^R \left(r - \frac{r^3}{R^2}\right) dr$$

$$\Rightarrow F = \frac{3}{16}\frac{GM^2}{R^2} \text{ Ans.}$$

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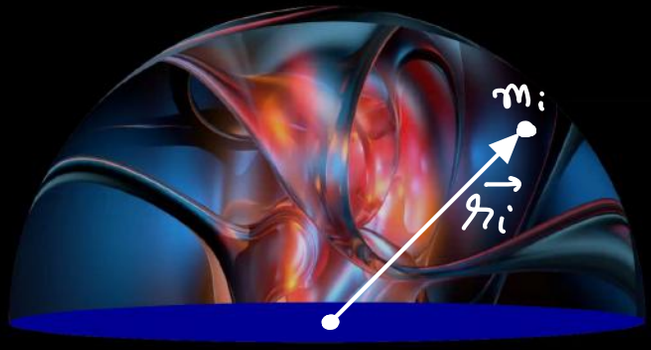
Alternate Method: Consider a particle of mass m_i located at a position vector $\vec{r}_i$ wrt centre of the hemisphere.

The force experienced by this particle is given as:

$$\vec{F}_i = \frac{-GM\vec{r}_i}{R^3} m_i$$



Thus net force is given as: $\vec{F} = \Sigma \vec{F}_i = -\frac{GM}{R^3}(\Sigma m_i \vec{r}_i) \times \frac{M/2}{M/2}$

Note that the mass of hemisphere is $M/2$ thus $\Sigma m_i \vec{r}_i / (M/2)$ is $\vec{r}_{cm}$ for hemisphere which is $\frac{3}{8}R\hat{j}$. Thus:

$$\boxed{\vec{F} = \frac{-3}{16}\frac{GM^2}{R^2}\hat{j} \text{ Ans}}$$

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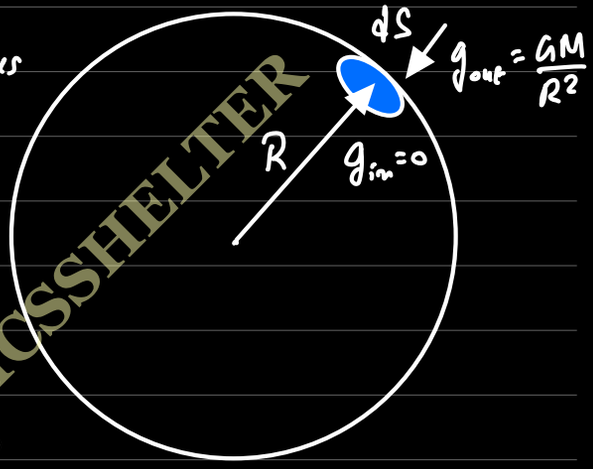
→ Note that due to Newton's third law, internal forces of the particles of hemisphere add up to zero and therefore we are left with only external forces. Thus we have the correct answer.

→ You may use similar techniques to find forces on other portions of solid sphere such as a wedge, octant, spherical cone and segment etc.

Gravitational Pressure On A Thin Spherical Shell :

Consider a spherical shell of mass M and radius R. We take a patch of area dS on its surface. Let the field contribution of this patch at a very near point be g_1 and due to rest of the shell be g_2. Then by shell theorems:



$$g_1 + g_2 = \frac{GM}{R^2} \text{ and } g_1 - g_2 = 0$$

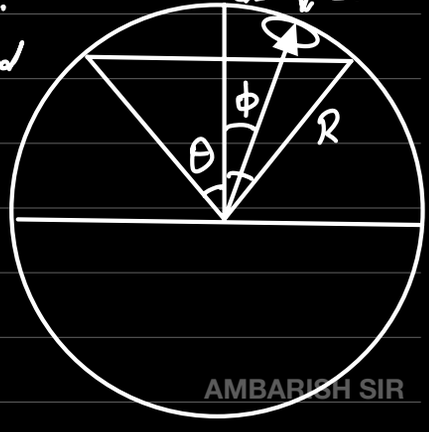
$\Rightarrow g_1 = g_2 = \frac{GM}{2R^2}$, Now let σ be the mass density per unit area, then, force experienced by the patch, $dF = \frac{GM}{2R^2}\sigma\, dS$

$\Rightarrow$ Pressure (attraction) experienced by the patch, $p = \frac{dF}{dS} = \frac{GM\sigma}{2R^2}$.

Now $M = \sigma \times 4\pi R^2$, thus $\boxed{p = 2\pi G\sigma^2}$

Example : Force On Cap Of A Spherical Shell :

Find the gravitational force on cap of a spherical shell with mass M and radius R. The semiverten angle of cap is θ.



Solution: Consider a patch of area dS, the force experienced by this patch is given as: $dF = 2\pi G\sigma^2 dS$,

Now vertical component of this force is:

$$dF\cos\phi = 2\pi G\sigma^2 dS\cos\phi.$$

Now recognise that $dS\cos\phi$ is nothing but shadow area of the patch in the equatorial plane,

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Thus $dF_{\text{sph}} = 2\pi G\sigma^2 \, dS_{\text{shadow}}$, now due to symmetry only vertical components of forces will add up, horizontal components of force will cancel out. Thus our required force is given as:

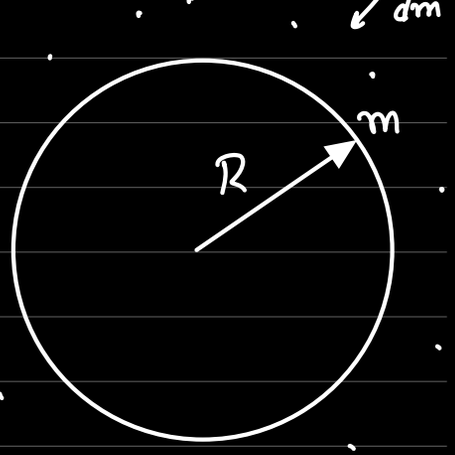
$$F = 2\pi G\sigma^2 \int dS_{\text{shadow}} = 2\pi G\sigma^2 \times \pi (R\sin\theta)^2$$

Thus required force is $2\pi^2 G\sigma^2 R^2 \sin^2(\theta)$ Ans.

GRAVITATIONAL SELF ENERGY: Negative of work done by gravitational forces in assembling a body bringing its infinitesimal elements from infinity is called its gravitational self energy. It also represents the change in Gravitational potential energy in process of assembling the body.

SELF ENERGY OF A SPHERICAL SHELL:

We wish to assemble a thin spherical shell of mass M and radius R, now suppose we have already assembled a mass m and now we further bring a mass dm from infinity to its surface. In this process, change in elemental PE:



$$dU = -\frac{Gm}{R}dm \quad \left(\because V = -\frac{Gm}{R}\right)$$

$$\Rightarrow U = -\frac{G}{R}\int_0^M m\,dm \Rightarrow \boxed{U = -\frac{GM^2}{2R}}$$

→ The expression for self energy can be used for finding the work required for expanding a shell, or obtaining the kinetic energy of particles of a contracting shell.

→ Self Energy of a point mass is infinity, however so long as point mass is not being split in a question, it is OK to take it as zero (the terms does not change so comes as it is on RHS & LHS of energy equation).

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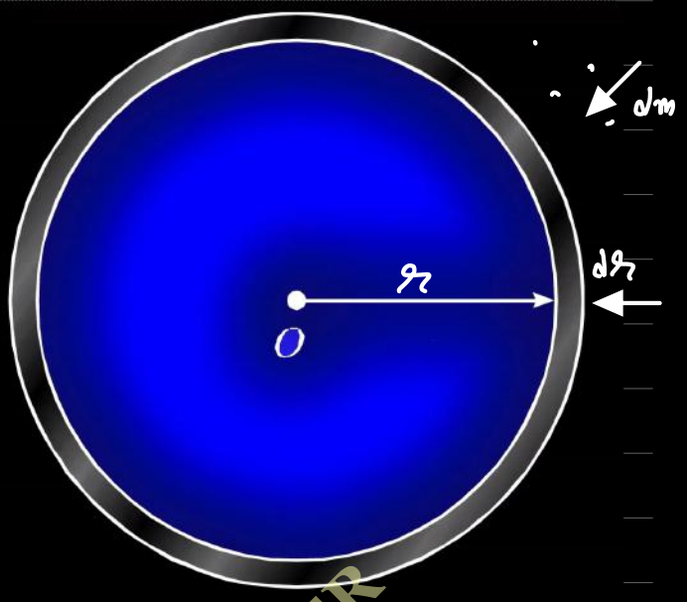
Self Energy Of A Solid Sphere:



We wish to assemble a solid sphere of mass M and radius R, now suppose we have already collected a mass m up to a radius r and we now bring another mass dm from infinity to assemble around it. In this process, elemental change in Gravitational PE:

$dU = -\frac{Gm}{r}dm$ $(\because dU = Vdm)$

Now, $m = \frac{M}{4/3\pi R^3} \times 4/3\pi r^3 = Mr^3/R^3 \Rightarrow r = R\left(\frac{m}{M}\right)^{1/3}$

now substituting r in dU and integrating:

$$dU = -\frac{Gm^{2/3}}{RM^{1/3}}dm \Rightarrow U = \frac{-GM^{1/3}}{R}\int_0^M m^{2/3}dm$$

$$\Rightarrow \boxed{U = \frac{-3GM^2}{5R}}$$

Total Gravitational Energy: Summation of all the self energies of bodies in a gravitational system and the interaction energies of various bodies is known as total gravitational energy.

For shells and spheres, the interaction energy can be written just as we did for point masses as long as spheres and shells do not enter each other.

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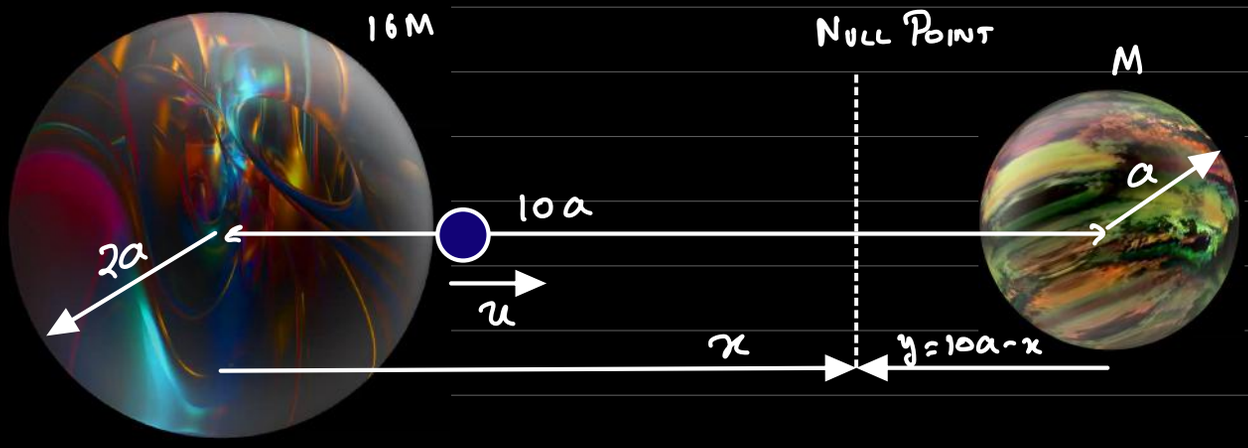
Example:

IITJEE 1996 Distance between the centres of two stars is $10a$. The masses of these stars are M and $16M$ and their radii a and $2a$ respectively. A body of mass m is fired straight from the surface of the larger star towards the smaller star. What should be its minimum initial speed to reach the surface fo the smaller star? Obtain the expression in terms of G, M and a.

Solution: As the body is receding away from a star, its velocity keeps decreasing until the gravitational attraction of the other star exceeds that of the launching star. This point is the point of maximum potential energy (Recall, $F = -\frac{dU}{dr}$, then zero force mean

an extremum of PE).



Let us first locate this null point. For null point,

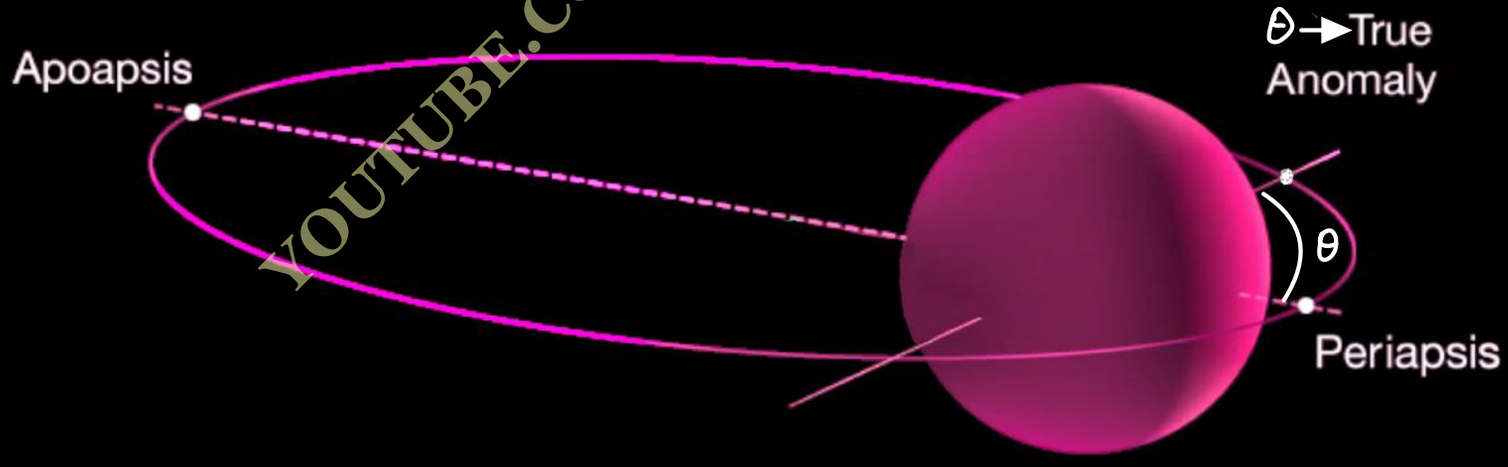
$$G \times 16M/x^2 = GM/(10a-x)^2 \Rightarrow x = 8a \Rightarrow y = 2a$$

Now in the limiting case, the projectile will reach the null point with zero velocity. Now conserving the mechanical energy:

$$\underbrace{-\frac{G \times 16M \times m}{2a}}_{\text{Due to larger}} - \underbrace{\frac{GMm}{8a}}_{\text{due to smaller}} + \frac{1}{2}mu^2 = -\frac{G \times 16M \times m}{8a} - \frac{GMm}{2a}$$

Solving for u, we get: $u = \sqrt{\frac{45GM}{4a}}$ Ans.

ELLIPTICAL ORBITS:



So far we have mathematically dealt only with circular orbits however in general the planetary orbits (or satellite orbits around planets) are elliptical, with the centre of the star serving as one of the foci of the ellipse. The point on the orbit nearest to the planet is called periapsis and the farthest point is called Apoapsis.

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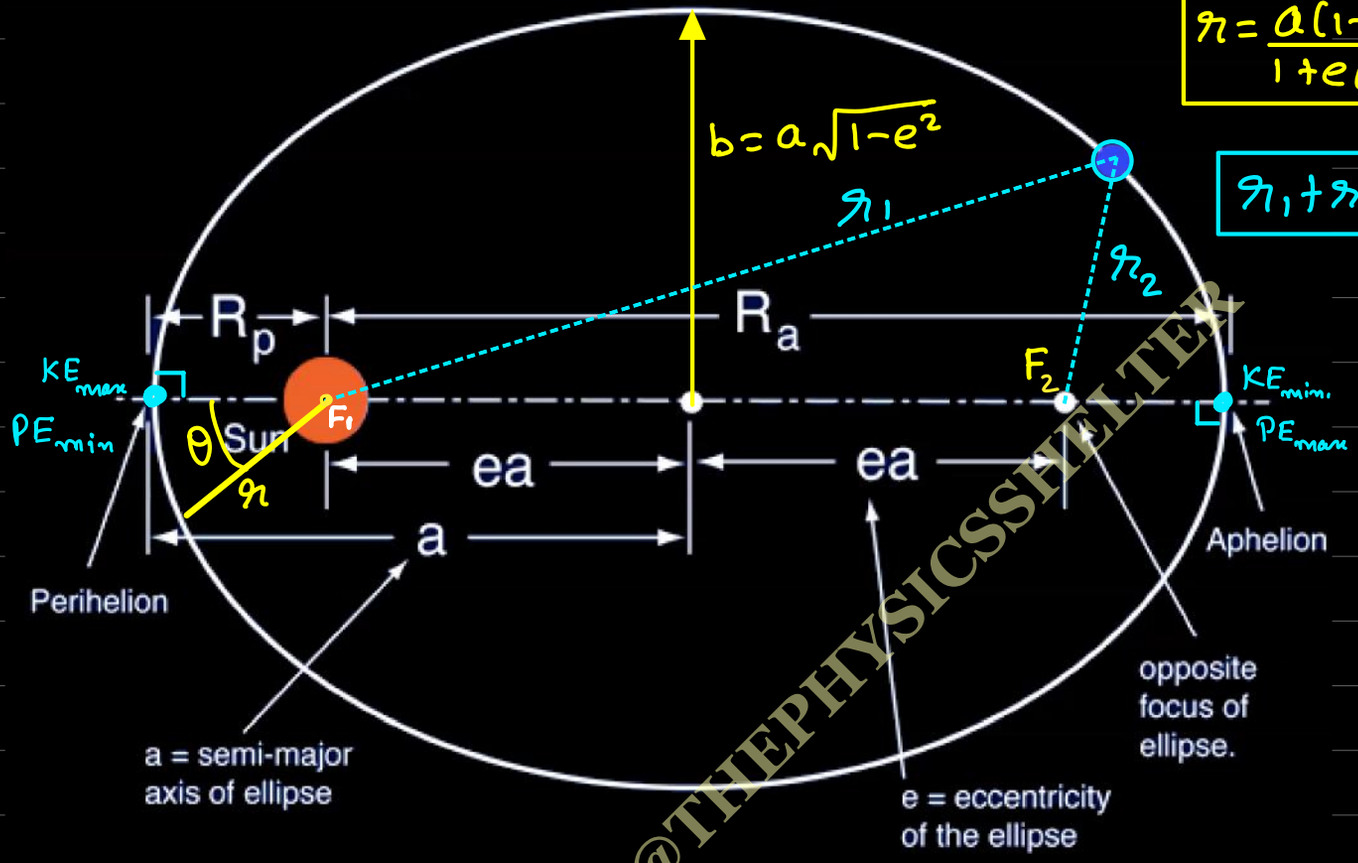
The angular position of the revolving body from the eccentricity vector (vector directed from apoapsis towards periapsis with magnitude equal to mathematical eccentricity) is called true anomaly.

The Geometry Of Elliptical Orbit:

$$r = \frac{a(1-e^2)}{1+e\cos\theta}$$

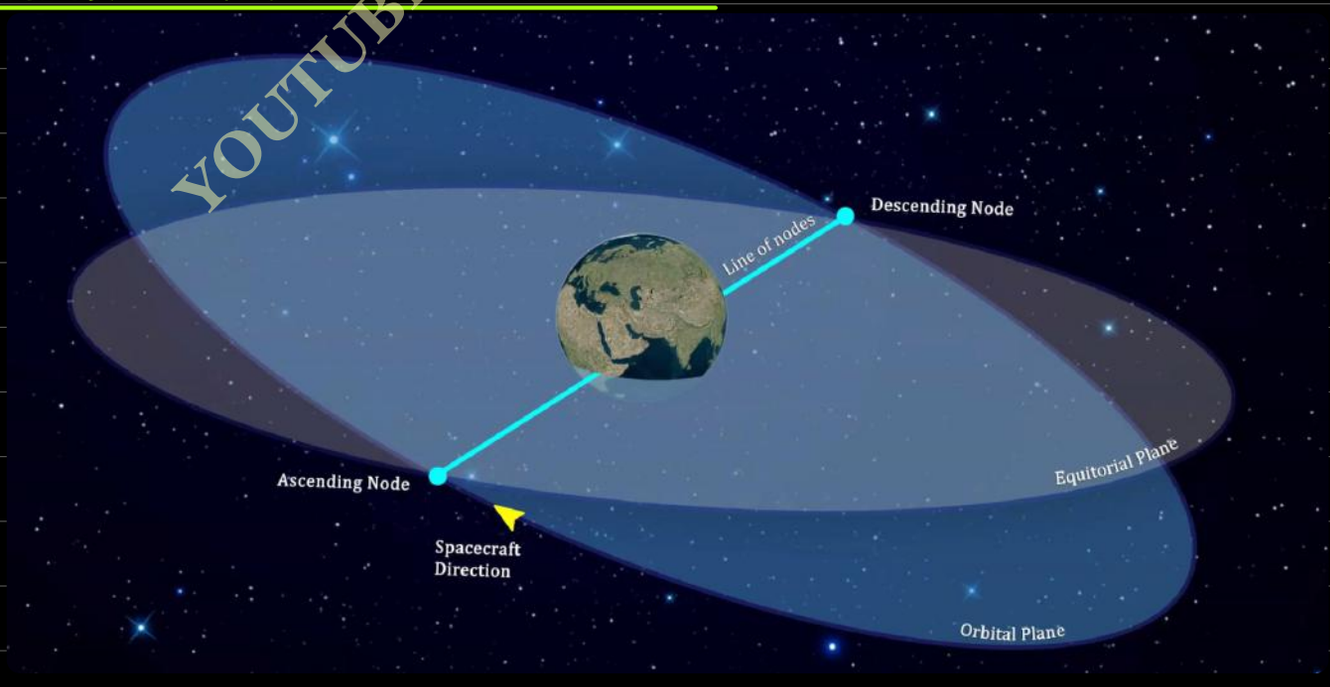
$$r_1 + r_2 = 2a$$



$$R_a = a(1+e)$$

$$R_p = a(1-e)$$

Ascending And Descending Nodes:



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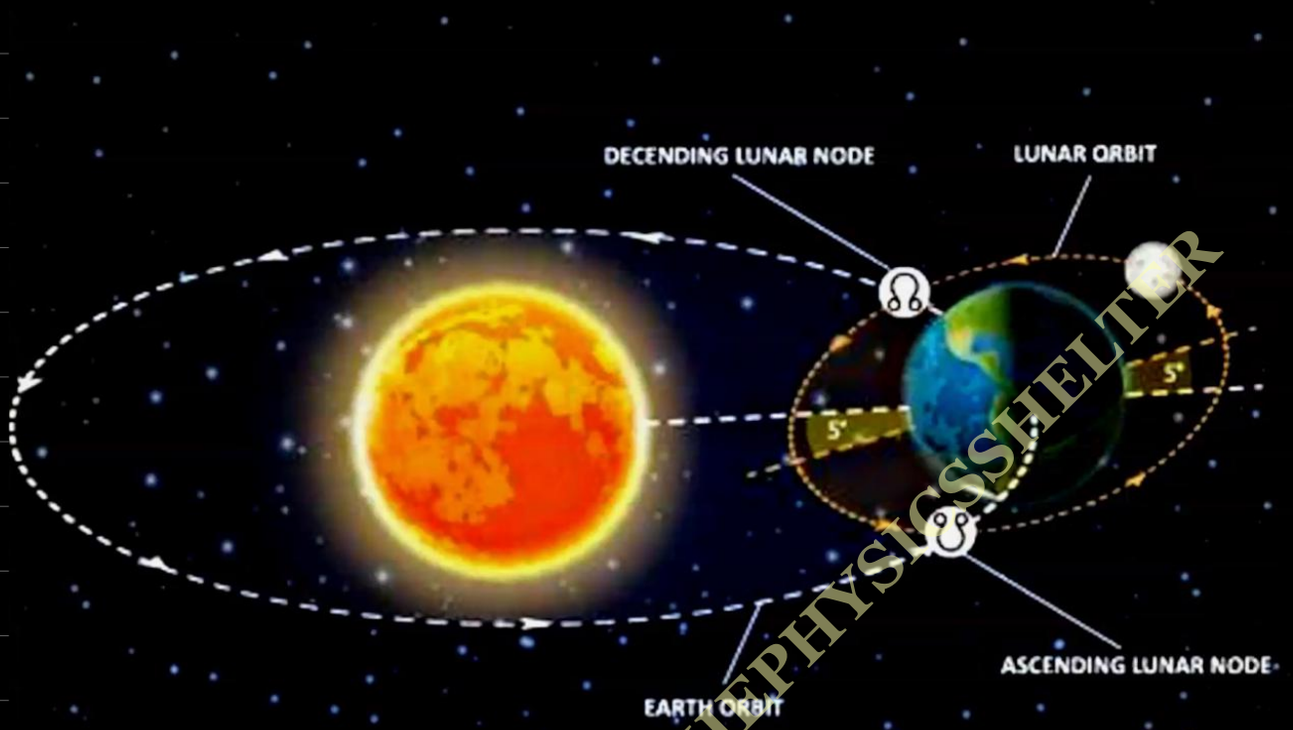
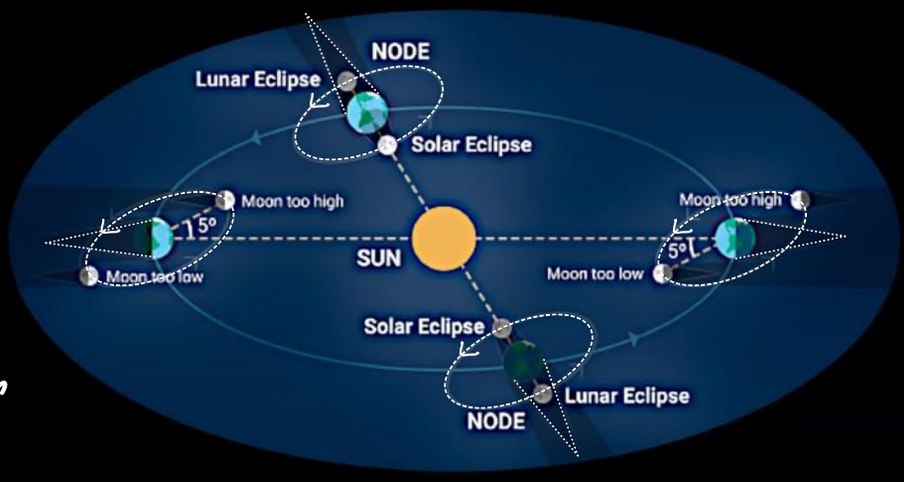
When a satellite crosses a particular plane the intersection points are called nodes. If the direction of crossing is northwards, it is called an ascending node if it is southwards we call it descending node.

Ascending & Descending Nodes On Ecliptic Circle



ASCENDING & DESCENDING LUNAR NODES

Understanding Eclipses (Inclined Orbit Of Moon) :



As clean from adjacent figure, the orbit of moon is not coplanar with ecliptic plane (plane of earth's revolution). Thus when earth is at certain locations, moon is never hidden from sun, whereas in certain positions of earth, moon can come between direct rays from sun to earth (Solar eclipse) or earth can come between direct rays from sun to moon (lunar eclipse). (Moon's orbit is inclined at 5° to the ecliptic plane).

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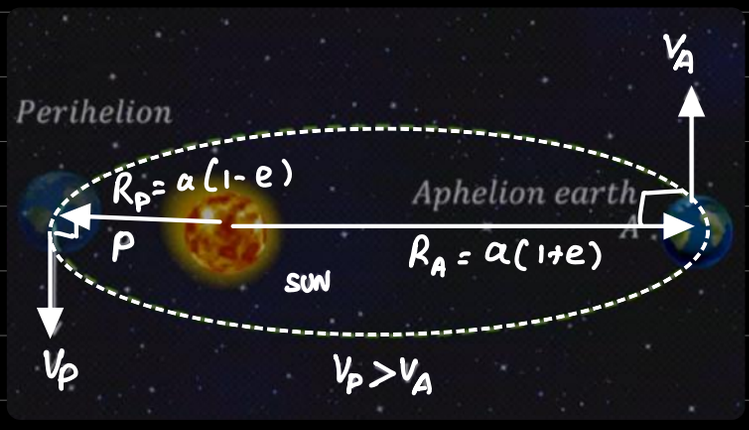
Perihelion & Aphelion Velocities In Elliptical Orbits:

Since gravitational force of sun on earth passes through centre of sun thus torque of gravitational force is zero, therefore angular momentum must be conserved. Thus:

$$m V_p a(1-e) = m V_A a(1+e) \quad (1)$$



Now by conservation of mechanical energy:

$$\frac{1}{2} m V_p^2 - \frac{GMm}{a(1-e)} = \frac{1}{2} m V_A^2 - \frac{GMm}{a(1+e)} \quad (2)$$

→ Note that in both the equations, m will cancel throughout thus V_p & V_A will be independent of mass of planet.

→ Here we have assumed sun to be stationary but actual motion would be around barycentre (centre of mass of sun/Planet system). But since sun is very massive we are ignoring this.

Now Solving (1) & (2)

$$V_p = \sqrt{\frac{GM}{a}\left(\frac{1+e}{1-e}\right)} \qquad V_A = \sqrt{\frac{GM}{a}\left(\frac{1-e}{1+e}\right)}$$

→ V_p is maximum velocity in the entire orbit because PE is minimum at perihelion

→ V_A is minimum velocity in the entire orbit because PE is maximum at Aphelion.

Mechanical Energy In Orbit:

$E = \frac{1}{2} m v^2 - \frac{GMm}{r}$, Since mechanical energy is conserved we can evaluate it at any point of our choice. Let us evaluate it at perihelion, thus @ $r = a(1-e)$, $v = \sqrt{\frac{GM}{a}\left(\frac{1+e}{1-e}\right)}$

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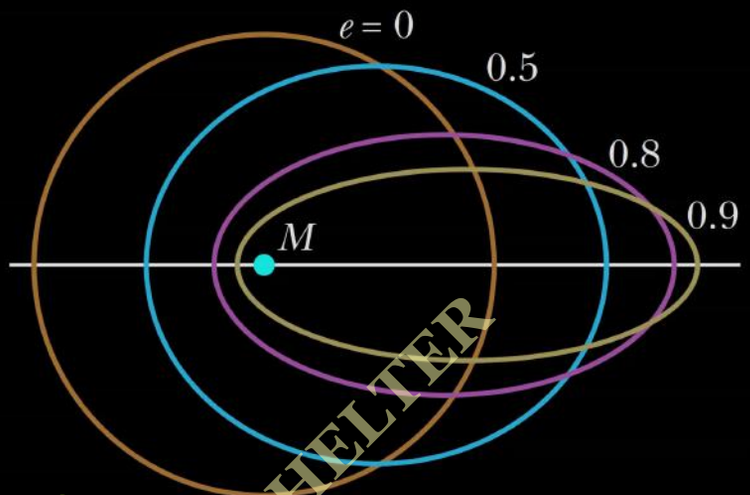
Thus $E = \frac{1}{2} m \left(\frac{GM}{a} \left(\frac{1+e}{1-e} \right) \right) - \frac{GMm}{a(1-e)}$

$= \frac{GMm}{a(1-e)} \left[\frac{1+e}{2} - 1 \right] = \frac{-GMm}{2a}$, Thus:

→ All orbits having same major axes have same total energy even if their eccentricities are different.

→ If velocity of projection is known, $2a$ is immediately known from the total energy expression.

→ If nearest distance is known, farthest distance can be found using $r_A = 2a - r_p$.

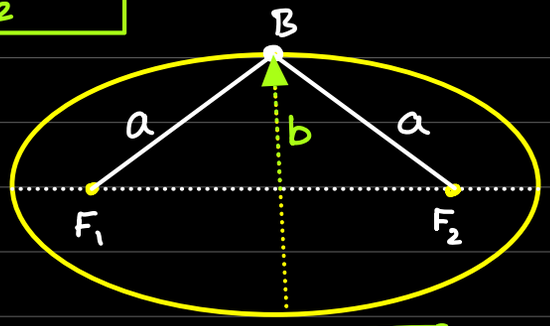


DIFFERENT e but same Energy E.

→ At the end point of minor axis, $\boxed{KE = |E| = \frac{1}{2}|PE|}$ this can be shown as follows:

$PE = -\frac{GMm}{a}$, $TE = -\frac{GMm}{2a} \Rightarrow KE = TE - PE$

$\Rightarrow KE = +\frac{GMm}{2a}$ thus $\boxed{KE = |E| = \frac{1}{2}|PE|}$



$b = a\sqrt{1-e^2}$

→ Velocity at end point of minor axis is:

$\boxed{V_B = \sqrt{\frac{GM}{a}}}$ this can be easily see as: $\frac{1}{2} m V_B^2 = \frac{GMm}{2a}$ (i.e KE)

thus $V_B = \sqrt{GM/a}$.

→ Distance of end points of minor axis from centre of sun is a and vice versa i.e if distance of a point in the orbit is a (i.e semimajor axis) then it must be the end point of minor axis.

→ If a projectile is launched with orbital velocity in arbitrary direction then the launch point is an end point of the minor axis. This fact can be easily proven as follows:

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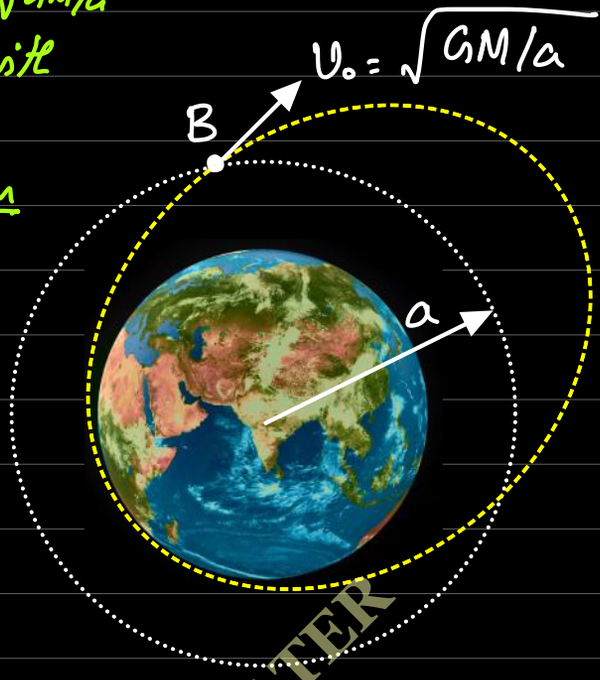
Consider a projectile launched with $v = v_0 = \sqrt{GM/a}$ as shown in the figure from a point with $r = a$. The energy of this projectile is:

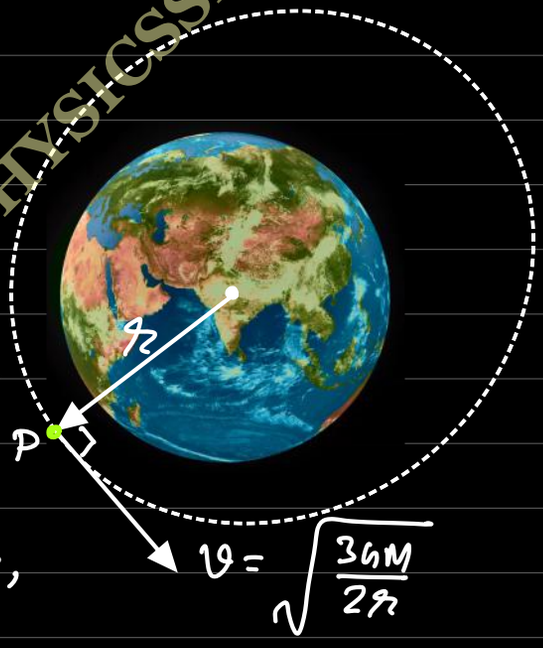
$$E = \frac{1}{2}mv_0^2 - GMm/a = \frac{GMm}{2a} - \frac{GMm}{a}$$

$\Rightarrow E = \frac{-GMm}{2a}$ $\Rightarrow$ Major axis is $2a$,



and launch point is at a distance of a from centre of earth (i.e focus), thus point B must be an end point of the minor axis.

EXAMPLE: A projectile is launched with speed $\sqrt{3GM/2r}$ from a distance r measured from centre of earth. The launching is perpendicular to the radius r. Find the maximum distance of projectile from centre of earth. Also find eccentricity of orbit and minor axis.



Solution: Since $v \perp r$, the launch point must be either a perigee or an apogee. Further since $v_0 < v < v_e$, therefore it must be the perigee.

Also $R_P + R_A = 2a$ thus $R_A = 2a - R_P$. Thus if we know the major axis, our job is done. Now total energy of our projectile:

$$E = -\frac{GMm}{r} + \frac{1}{2}m \times \frac{3GM}{2r} = -\frac{GMm}{4r} = \frac{-GMm}{2a}$$

$\Rightarrow 2a = 4r$ thus farthest distance : $R_A = 2a - R_P = 4r - r = 3r$

Thus farthest distance is $3r$. Ans.

Now farthest distance $= a(1+e) = 2r(1+e) = 3r \Rightarrow e = \frac{1}{2}$ Ans.

Now $b = a\sqrt{1-e^2} = 2r\sqrt{1-1/4} = \sqrt{3}r$ thus $2b = 2\sqrt{3}r$ Ans.

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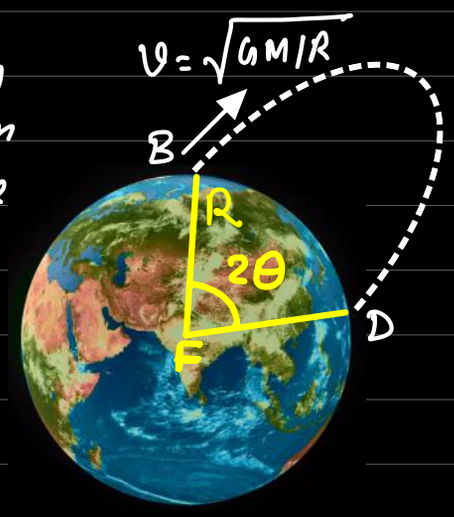
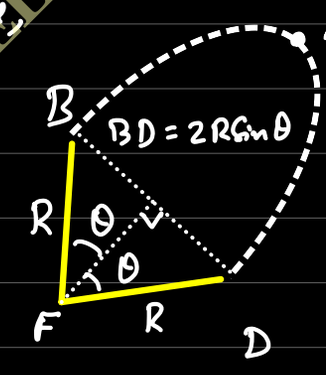
EXAMPLE: A projectile is launched from north pole so as to land after covering an angle 2θ along longitudes as shown in the figure. The launch speed is $\sqrt{GM/R}$. Find (a) Maximum distance of projectile from centre of earth (b) eccentricity of the orbit.



Solution: Since launch speed is equal to orbital speed, the launch point must be end point of minor axis and so also the landing point. Thus BD is minor axis of the elliptical orbit. Thus from adjacent figure,

$BD = 2b = 2R\sin\theta$, also $a = R$ (∵ distance of end point of minor axis is equal to semi major axis). Now $b = a\sqrt{1-e^2}$

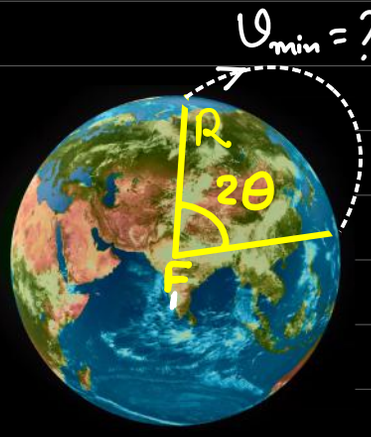
$\Rightarrow R\sin\theta = R\sqrt{1-e^2} \Rightarrow e = \cos\theta$



Now $FA = a(1+e) = R(1+\cos\theta)$, thus maximum height above surface: $h_{max} = FA - R = R\cos\theta$

thus $h_{max} = R\cos\theta$ Ans.

EXAMPLE: A projectile is to be launched from North pole to land at a latitude 2θ apart as shown in the figure, find the minimum launching speed required to accomplish this.



Solution: We shall use the following facts for solving the problem:

① Total energy of orbit is $-GMm/2a$, thus least velocity orbit is also the orbit with smallest major axis.

② In any ellipse, the sum of distances of any point to the two foci is $2a$

③ One of the Foci of the elliptical orbit is centre of the planet.

④ The other focus of the ellipse lies on the line joining

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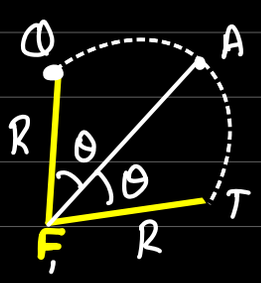
F_1 to Apogee.

(5) Line joining F_1 to A bisects angle subtended by equidistant points on F_1.



Using these facts our problem reduces to locating a point F_2 on line F_1A such that $F_1O + F_2O$ is minimum (because $F_1O + F_2O = 2a$ and we are looking for orbit with smallest a).

For smallest distance OF_2, we just draw a perpendicular from O on AF_1.

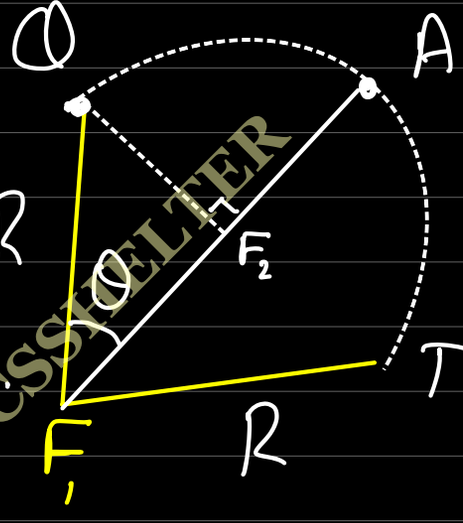
Thus $OF_2 = R\sin\theta$, now total major axis $2a = OF_1 + OF_2 = R + R\sin\theta$

Thus $2a = R(1+\sin\theta)$. Now we can easily find the corresponding launch velocity using mechanical energy conservation.

Thus:

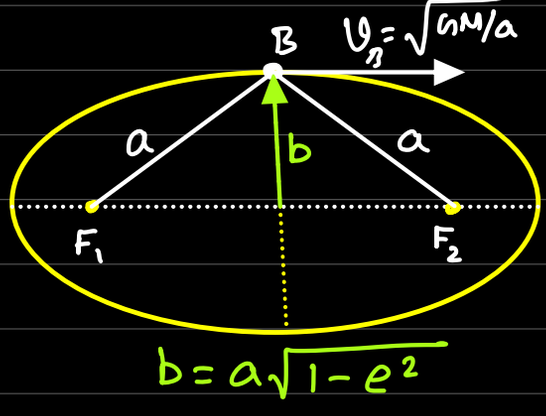
$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{-GMm}{R(1+\sin\theta)}$$

$$\Rightarrow v_{min} = \sqrt{\frac{2GM}{R}}\sqrt{\frac{\sin\theta}{1+\sin\theta}} \quad \text{Ans.}$$



Angular Momentum Of A Satellite:

Recall that speed of a satellite at the end point of minor axis is $v_B = \sqrt{GM/a}$ also angular momentum $= r_\perp mv$, thus as can be seen clearly from adjacent figure, angular momentum is given by:



$$L = b \times mv_B = mb\sqrt{\frac{GM}{a}}$$

Thus $\boxed{L = mb\sqrt{\frac{GM}{a}}}$

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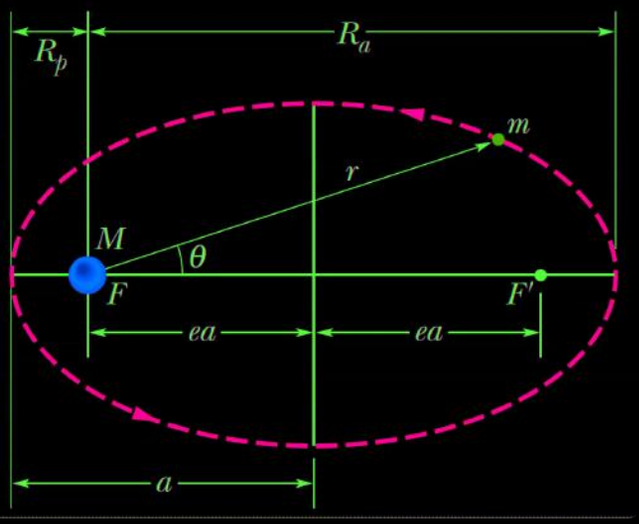
KEPLER'S LAWS:

FIRST LAW: LAW OF ORBITS:



All planets move in elliptical orbits with Sun as one of the Foci. We shall not be doing a complete proof of law of orbits (Involves a lot of Bloodbath!) however we shall see a strategy for solving orbital trajectories in a general central force field (Such as gravity of Electrostatic force on a particle tied to a spring).

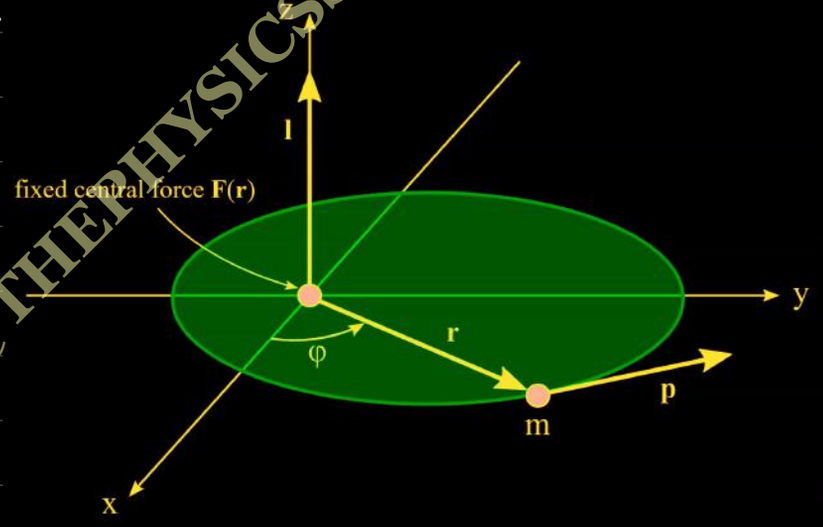
SOLVING ORBITAL TRAJECTORIES (CENTRAL FORCES):

Any typical central force problem is governed by following conservation laws:

① Conservation of angular momentum.

② Conservation of mechanical energy.

Let general coordinate of particle and centre of the force be (r, ϕ) as shown in figure. Then we have:



$$L = mr^2\dot{\phi} = \text{Constant} \quad \text{(I) (Angular Momentum Conservation)}$$

Also: $U(r) + \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\phi}^2 = E = \text{Constant.}$ (II)

Now these are simultaneous differential equations in r, ϕ and t, in a typical problem we generally know the initial conditions, thus L and E are known. However before we may proceed for any integration we should have a differential equation only in two variables, so we must carefully eliminate one of the variables.

Now From equation (I),

$$r^4\dot{\phi}^2 = L^2/m^2, \text{ using this in (II)}$$

we get:

$$U(r) + \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} = E \quad \text{(III)}$$

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Now recognise that suddenly in equation (III) we are only left with variables r and t. If we define an effective potential Energy $U_{eff}(r)$ as: $\boxed{U_{eff}(r) = U(r) + L^2/2mr^2}$, our equation (III) looks like the problem of a particle in a one dimensional potential on r axis, we get:

$$\boxed{U_{eff}(r) + \frac{1}{2}m\dot{r}^2 = E} \quad (IV)$$

Now equation (IV) can be integrated with appropriate initial conditions to get $r(t)$. Once we have $r(t)$, we can find $\phi(t)$ using equation (I) i.e:

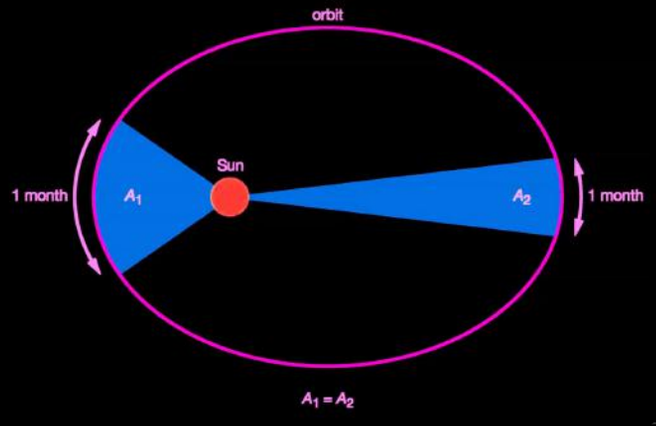
$L = mr^2\dot{\phi}$, this will now be a differential equation only in ϕ and t and thus we will get $\phi(t)$.

Once we have $r(t)$ and $\phi(t)$ we can eliminate t to get polar equation of the trajectory as $r(\phi)$.

The solution to the trajectory under Gravitational forces is quite cumbersome though very much doable within JEE Mathematics. If you wish to see the complete proof you can refer to David Morin's chapter on central forces (Introduction to Classical Mechanics).

KEPLER'S SECOND LAW: LAW OF AREAS:

The line connecting planet to sun sweeps out equal areas in equal intervals of time. (or dA/dt = Constant).



Kepler's Law of Areas is based on conservation of angular momentum and can be proved as follows.

Consider the planet at point P with position vector $\vec{r}$ and instantaneous velocity $\vec{v}$, then displacement of planet in a time dt is given as $d\vec{s} = \vec{v}\,dt$. Now altitude of the triangular swept area on $\vec{r}$ is $v\,dt\sin\theta$ thus area swept is given as:

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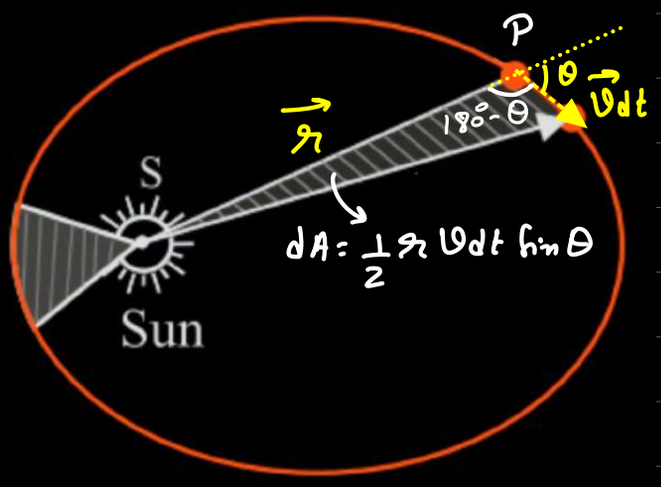
$dA = \frac{1}{2} r\, v\, dt \sin\theta$, alternatively if we take area as vector, recall from discussion on vectors that area vector of any triangle is simply $\frac{1}{2}$ x cross product of its sides. Thus:

$$d\vec{A} = \frac{1}{2}\vec{r} \times \vec{v}\, dt$$



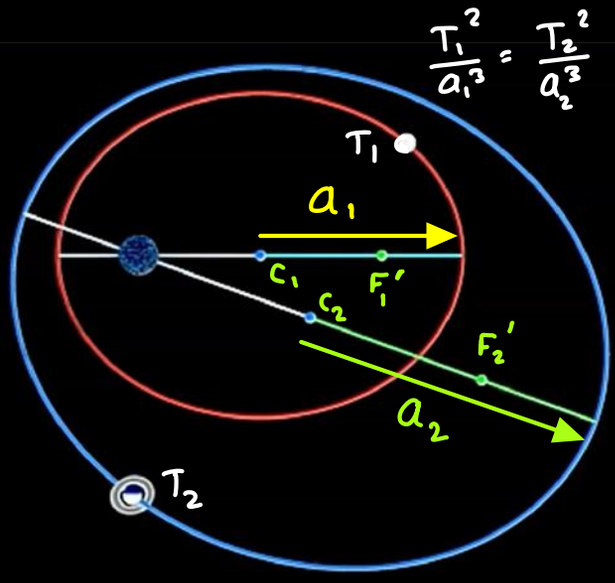
Here direction of area vector is in to the page. (Curl fingers of your right hand in the sense of revolution, thumb now gives direction of area vector.)

Thus we can also write elemental area vector as:

$$d\vec{A} = \frac{1}{2}\vec{r} \times m\vec{v}\, dt \times \frac{1}{m} = \frac{\vec{L}}{2m} dt,$$

Thus $\boxed{\frac{d\vec{A}}{dt} = \frac{\vec{L}}{2m}}$ this must be constant because by law of conservation of angular momentum, $\vec{L}$ must be a constant. QED. (One can readily see that $\vec{L}$ is along $d\vec{A}$).

KEPLER'S THIRD LAW: The square of time period of any planet is proportional to cube of semi major axis of its orbit. This law can also be easily proved using the fact that $dA/dt = L/2m$ and that $L = mb\sqrt{\frac{GM}{a}}$. The proof is as follows:



Area of Ellipse $A = \pi ab$

time for completing the orbit T = Area / Areal velocity $= \pi ab/(L/2m)$

Thus $T = \frac{\pi ab \times 2m}{mb\sqrt{\frac{GM}{a}}} = 2\pi\sqrt{\frac{a^3}{GM}} \Rightarrow \boxed{T^2 = \frac{4\pi^2}{GM} a^3}$

QED.

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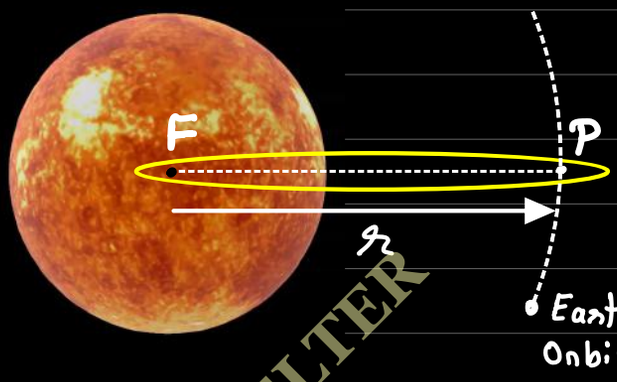
Example: Irodov 1.203:

A small body starts falling onto the Sun from a distance equal to the radius of the Earth's orbit. The initial velocity of the body is equal to zero in the heliocentric reference frame. Making use of Kepler's laws, find how long the body will be falling.

Solution: The path of the body is a line segment which is a case of degenerate ellipse with end points as the Foci. Thus if R is the semimajor axis of earth's orbit, for the line segment orbit, the semi-major axis is $R/2$. Now using Kepler's law if T_B is round trip time of the body then:



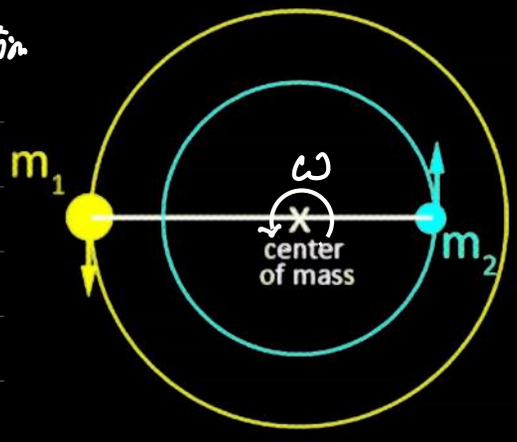
$$T_B^2/(1\,\text{Year})^2 = (R/2)^3/R^3 = 1/8$$

$\Rightarrow$ $T_B = \sqrt{1/8}$ years, however we want just the time of falling in to sun and not the round trip time thus our required answer must be half of this thus:

$$T_f = \frac{T_0}{2} = \frac{1}{\sqrt{32}}\,\text{Year} \approx 65\,\text{days}.$$

Binary Stars (Circular Orbits):

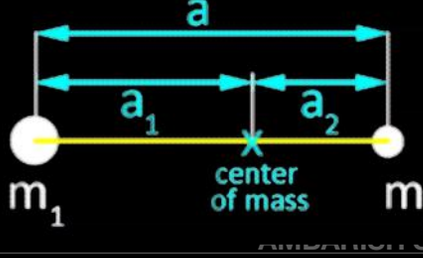
Two stars can execute circular motion under action of mutual gravitation about common centre of mass we shall prove the following results:



(1) $\boxed{T = \frac{4\pi^2 a^3}{GM}}$ where $\boxed{M = m_1 + m_2}$

(2) $\boxed{E = -\frac{GM\mu}{2a}}$ where $\boxed{\mu = \frac{m_1 m_2}{m_1 + m_2}}$

(3) $\boxed{L = \mu b\sqrt{\frac{GM}{a}}}$ where $\boxed{b = a}$



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Here a is distance between m_1 and m_2.

CM divides line joining the particles in inverse ratio of masses

thus $a_1 = \frac{m_2}{m_1+m_2}\, a$ and $a_2 = \frac{m_1}{m_1+m_2}\, a$

Now balancing centrifugal and Gravitational forces on m_1:

$$m_1\omega^2 a_1 = \frac{G m_1 m_2}{a^2} = m_1\omega^2 \times \frac{m_2}{m_1+m_2}\, a$$

$$\Rightarrow \omega^2 = \frac{G(m_1+m_2)}{a^3} = \left(\frac{2\pi}{T}\right)^2 \Rightarrow \boxed{T^2 = \frac{4\pi^2}{GM}\, a^3}$$ where $M = m_1+m_2$. QED.

Now KE in CM frame, $KE = \frac{1}{2}\mu V_{rel}^2 = \frac{1}{2}\mu(\omega a)^2 = \frac{1}{2}\mu a^2\omega^2$

Thus total energy: $E = \frac{1}{2}\mu a^2\omega^2 - \frac{G m_1 m_2}{a}$

$$\Rightarrow E = \frac{1}{2}\mu a^2 \cdot \frac{GM}{a^3} - \frac{GM\mu}{a} = -\frac{GM\mu}{2a} \Rightarrow \boxed{E = -\frac{GM\mu}{2a}}$$ QED.

Now the two stars form a momentum couple, thus their angular momentum is independent of choice of point. Also for two body system $p = \mu V_{rel} = \mu\omega a$, thus angular momentum about CM:

$$L = \mu\omega a \times a_1 + \mu\omega a \times a_2 = \mu\omega a(a_1+a_2) = \mu\omega a^2$$

Thus $L = \mu a^2\sqrt{\frac{GM}{a^3}} = \mu a\sqrt{\frac{GM}{a}} = \mu b\sqrt{\frac{GM}{a}}$ where $b = a$ (QED)

→ Even though so far we have proved these results for circular orbits, same results can also be used for finding time periods and eccentricities of elliptical orbits.

→ 'a' can be found using $E = -GM\mu/2a \Rightarrow \boxed{a = -\frac{GM\mu}{2E}}$

→ 'b' can be found using $L = \mu b\sqrt{\frac{GM}{a}}$

→ 'e' can be found using $b = a\sqrt{1-e^2}$

→ Now maximum and minimum distances between stars can be found as $r_{min} = a(1-e)$ and $r_{max} = a(1+e)$.

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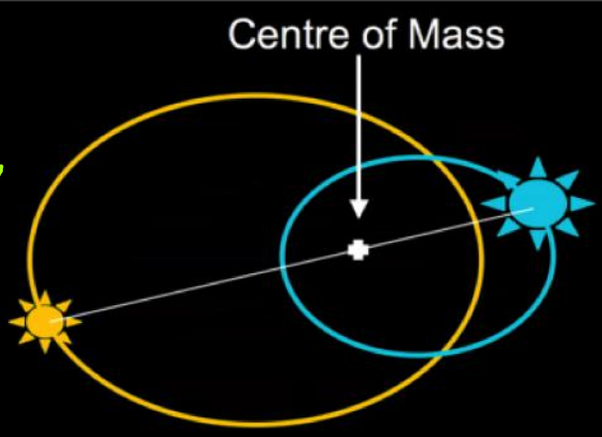
→ In case of elliptical orbits, the two stars will execute similar ellipses with common focus at centre of mass, and linear dimensions being inversely proportional to masses of stars. The individual maximum and minimum distances from CM can be calculated by dividing their mutual maximum/minimum distances in inverse ratio of masses.



→ For a bound orbit, Total Energy of the system in CM frame must be negative otherwise bodies will escape to infinity. If the energy in CM frame is zero, particles will escape to infinity and will have zero KE there, if the system has a positive energy, even at infinity the bodies will have a finite KE. ($E<0 \Rightarrow$ Elliptical orbit, $E=0 \Rightarrow$ Parabolic trajectory, $E>0 \Rightarrow$ Hyperbolic Trajectory).

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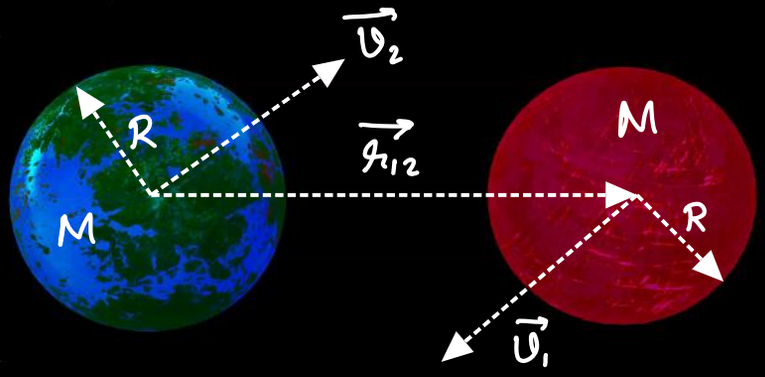
EXAMPLE: PATHFINDER CHECK 1:

Two asteroids each of mass of mass M and radius R are moving in free space. At an instant, when their velocity vectors are $\vec{v}_1$ and $\vec{v}_2$ respectively, position vector of the second with respect to the first is $\vec{r}_{12}$. Determine the condition under which they will collide.

Solution: For collision, minimum centre to centre distance must be less than $2R$. Thus in limiting case:

$$a(1-e) = 2R \quad (1)$$



Now $\mu = M/2$, $M \to 2M$ ($\because m_1 + m_2$).

Now total Energy $\frac{1}{2}\left(\frac{M}{2}\right)v_{21}^2 - \frac{GM^2}{r_{12}} = -\frac{G \times (2M)(M/2)}{2a}$ (2)

Angular Momentum:

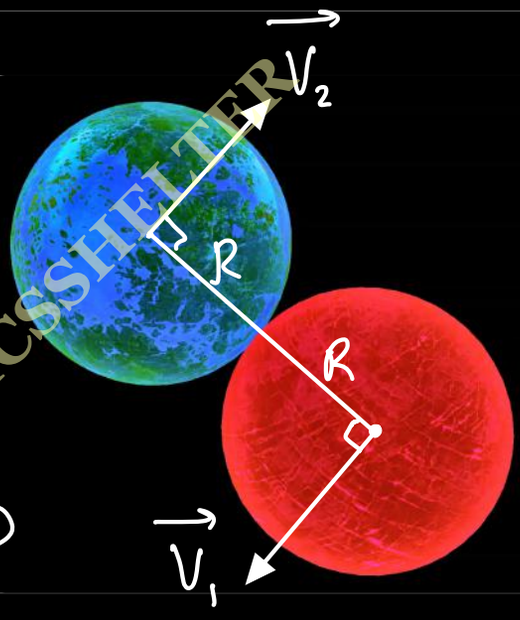
$$\Rightarrow |\vec{r}_{12} \times \vec{v}_{12}| \times \frac{M}{2} = \frac{M}{2} b \sqrt{\frac{G \times 2M}{a}} \quad \Rightarrow b = \sqrt{\frac{a}{2GM}} |\vec{r}_{12} \times \vec{v}_{12}| \text{ (3)}$$

Also: $b = a\sqrt{1-e^2}$ (4)

Now eliminating a, b and e from (1), (2), (3) & (4):

$$\frac{|\vec{r}_{12} \times \vec{v}_{21}|}{16R^2} > \frac{v_{21}^2}{4} + GM\left[\frac{1}{2R} - \frac{1}{r_{12}}\right] \quad (> \text{ sign for collision})$$

Alternate Method: In the limiting case, the two spheres will just be grazing past each other such that the relative velocity of the two spheres is in the tangent plane at the contact point. Now by mechanical energy conservation:



$$\frac{1}{2}\frac{M}{2}v_{21}^2 - \frac{GM^2}{r_{12}} = \frac{1}{2}\frac{M}{2}V_{21}^2 - \frac{GM^2}{2R} \text{ (1)}$$

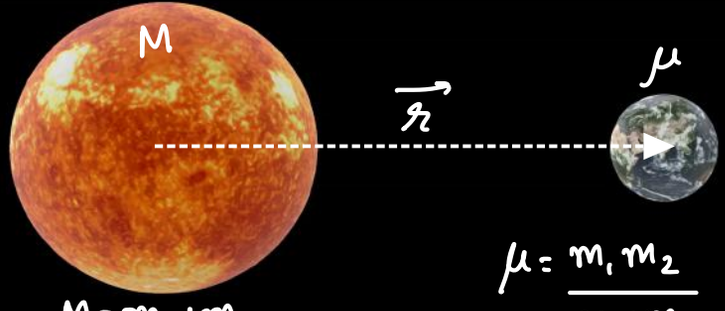
Using angular momentum conservation:

$$\frac{M}{2} \times V_{21} \times 2R = \frac{M}{2}|\vec{r}_{12} \times \vec{v}_{21}| \text{ (2)}$$

eliminating V_{21} from the two equations we will get the same desired result. (This system is easier to solve than the previous one).

Proof Of Reduced Mass & Effective Fixed Star Approach:

So far we have been using the approach of effective fixed star and reduced mass without proof. Now we shall prove that the differential equation governing this system is exactly same as the differential equation governing the separation vector between binary stars.



$M = m_1 + m_2$ (FIXED STAR)

$\mu = \frac{m_1 m_2}{m_1 + m_2}$ (REDUCED MASS PLANET)

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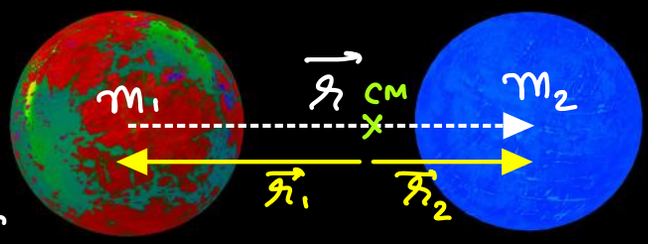
Using Newton's second law we can write the differential equation of the system as:

$$-\frac{GM\mu}{r^2}\hat{r} = \mu\ddot{\vec{r}} \quad (1)$$

Now consider the binary star system moving about combined CM. Let position vector of star 2 wrt star 1 be $\vec{r}$. Recall that momentum of any body in CM frame for a two body system is given as: $\vec{p} = \mu\vec{v}_{rel}$ where $\vec{v}_{rel}$ is the velocity of one body relative to the other. Now writing Newton's second law in the form $\vec{F} = d\vec{p}/dt$ for star 2:



$$-\frac{Gm_1m_2}{r^2}\hat{r} = \frac{d}{dt}(\mu\vec{v}_{rel}) = \frac{d}{dt}(\mu\dot{\vec{r}}) = \mu\ddot{\vec{r}}$$

thus $-\frac{Gm_1m_2}{r^2} \times \left(\frac{m_1+m_2}{m_1+m_2}\right)\hat{r} = \mu\ddot{\vec{r}}$, thus:

$$-\frac{GM\mu}{r^2}\hat{r} = \mu\ddot{\vec{r}} \quad (2)$$ this is exactly the same equation as the one obtained for reduced mass planet revolving around an effective fixed star and therefore its solution must also exactly same as the one for the equivalent fixed star system.

→ Realise here $\vec{r}$ is the position of one star wrt to the other accordingly if we want individual positions wrt CM we need to divide $\vec{r}$ in the inverse ratio of masses with the appropriate sign. Thus: $\vec{r}_2 = \frac{m_1}{m_1+m_2}\vec{r}$ and

$$\vec{r}_1 = -\vec{r}\left(\frac{m_2}{m_1+m_2}\right).$$

Note this modelling also holds correct for kinetic Energy, Potential Energy and angular momentum as well (when considered for a

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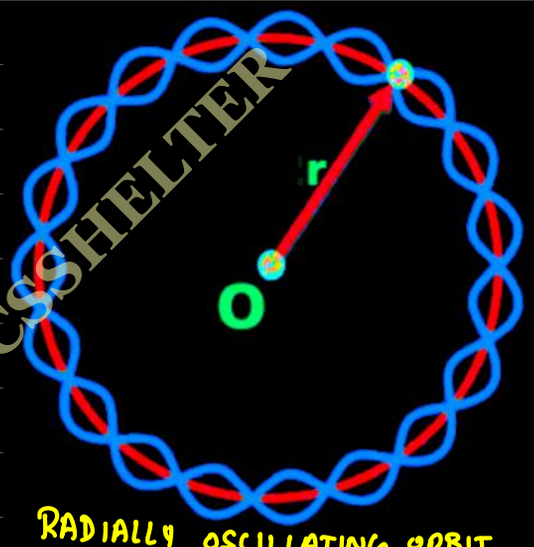
system as a whole). Thus: $-\frac{GM\mu}{r}$ is the PE for both systems, $\frac{1}{2}\mu v_{rel}^2$ is KE for both systems and $\vec{r} \times \mu \dot{\vec{r}}$ is the angular momentum for both systems and $-\frac{GM\mu}{r^2}\hat{r}$ is the Gravitational force for both systems. (Perfect analogy for the systems).

ORBIT STABILITY UNDER CENTRAL FORCES:

Suppose a particle is executing circular motion in an orbit of radius r under action of a central force, whose potential energy varies as $U(r)$, then recall that we can convert the problem of solving for its orbit in to one dimensional problem: $U_{eff}(r) + \frac{1}{2}m\dot{r}^2 = E$



RADIALLY OSCILLATING ORBIT.

where $U_{eff} = U(r) + L^2/2mr^2$,

Now suppose the orbiting body is given some kick with a component in radial direction and we wish to know whether the particle will spiral out to infinity or spiral in to the centre or keep oscillating between a certain maximum and minimum radius. This problem is just like considering stability of a particle in one dimensional potential. Thus, for stable orbit, the particle must be in the valley of $U_{eff}(r)$ curve. Therefore for stability:

$$\boxed{U'_{eff}(r) = 0}, \qquad \boxed{\frac{d^2U_{eff}}{dr^2} > 0}$$

Also for sufficiently small kicks, the time period of radial oscillations can be found using:

$$\boxed{T = 2\pi\sqrt{\frac{m}{U''_{eff}(r)}}}$$ if Simple Harmonic.

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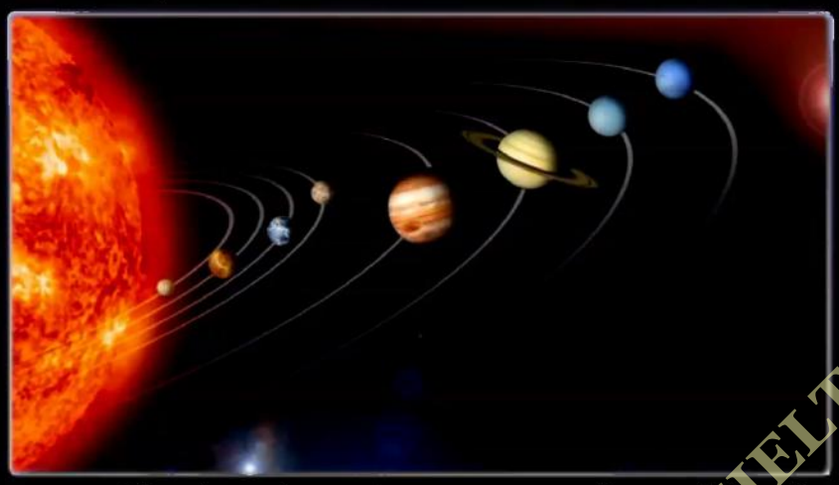
(Simple harmonic motion can be checked by looking at the taylor/binomial expansion terms near equilibrium position. If dominant quadratic term exists in $U_{eff}(r)$, then the motion is SHM).

EXAMPLE:

INPHO 2013 PROBLEM 6| RADIAL OSCILLATIONS UNDER CENTRAL FIELDS



(I) As you know that planets move in elliptical orbits under action of gravity of Sun. Suppose we assume that orbit is circular of radius r_o and planet is slightly disturbed from its position such that its position is $r = r_o+\delta$ where $\delta/r_o<<1$. Show that planet will oscillate simple harmonically around mean position r_o. Obtain time period (T_r) for these radial oscillations. Given mass of sun is M and gravitational constant is G.

(II) For this part we assume that planet moves under the force expressible as a power of r as $F=-cr^n$ where c ($c > 0$) is a constant. For what values of n a stable orbit is possible? (Orbit may not necessarily be nearly circular).

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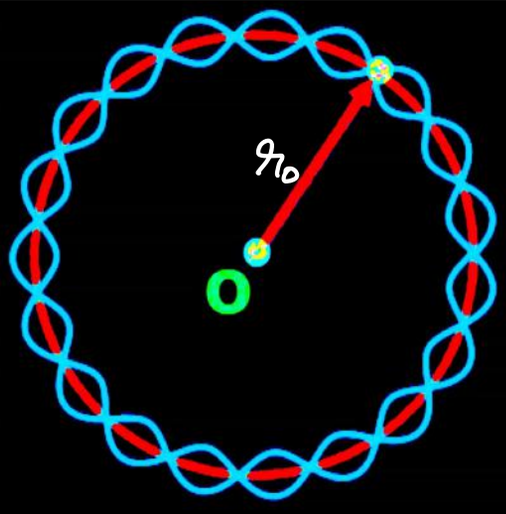
Solution: (a) Recall angular momentum:

$$L = mb\sqrt{\frac{GM}{a}}, \text{ for}$$



circular orbits, $b=a=r_o$ where r_o is orbit radius, thus $L = m\sqrt{GMr_o}$

Now

$$U_{eff} = -\frac{GMm}{r} + \frac{L^2}{2mr^2}$$

$$U'_{eff}(r) = \frac{GMm}{r^2} + \frac{L^2}{2m}(-2r^{-3})$$

$$U''_{eff}(r) = \frac{-2GMm}{r^3} + \frac{3L^2}{mr^4} \Rightarrow U''(r_0) = \frac{GMm}{r_0^3}$$

Now time period of oscillations $T = 2\pi\sqrt{\frac{m}{U''_{eff}(r_0)}} = 2\pi\sqrt{\frac{r_0^3}{GM}}$ Ans.

(b) $F(r) = -Cr^n = -dU/dr \Rightarrow U(r) = Cr^{n+1}/n+1.$

$$U_{eff} = \frac{Cr^{n+1}}{n+1} + \frac{L^2}{2mr^2},$$

Now for equilibrium, $U'_{eff} = 0$ and for stability $U''_{eff} > 0$

Now, U'_{eff}:

$$Cr^n + \frac{L^2}{2m}(-2r^{-3}) = 0 \Rightarrow \frac{L^2}{m} = Cr^{n+3}$$

Now $U''_{eff} = nCr^{n-1} + \frac{3L^2}{mr^4} = nCr^{n-1} + \frac{3Cr^{n+3}}{r^4}$

$\Rightarrow U''_{eff} = C(n+3)r^{n-1} > 0 \Rightarrow n > -3$ Ans.

→ For a central force of the form $F(r) = -Cr^n$ if $n \leq -3$, the orbit is not stable. (For $n = -3$, the trajectories belong to a family of spirals known as Cotes' spirals in which depending on perturbation either $r \to \infty$ or $r \to 0$). For $n = -3$, all derivatives of U''_{eff} become zero thus perfectly neutral equilibrium like condition (Plateau of potential energy in one dimension), a slight push to a block on plateau will take it all the way to ∞ (or towards radial origin).

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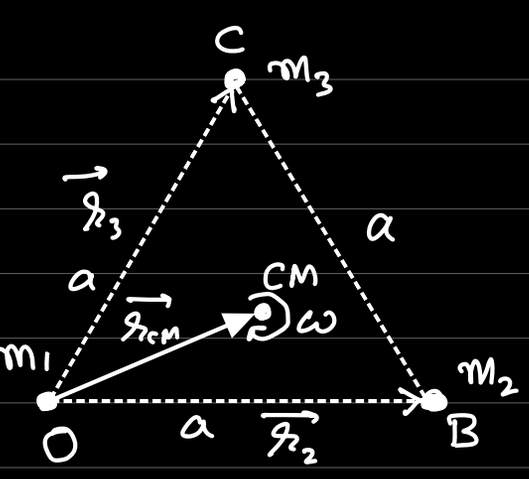
EXAMPLE : PATHFINDER : BUILD-9

Three material points of masses m_1, m_2 and m_3 are at the vertices of an equilateral triangle of side a. The system is rotating in free space in such a way that under the mutual gravitational interaction of the three particles, the system is neither expanding nor contracting. Find angular velocity ω of this rotation. Universal gravitational constant is G.

Solution: The system must be rotating about common centre of mass with particles executing circular motion. Now applying laws of motion to m_1:



$$\frac{Gm_1m_3}{a^3}\vec{r}_3 + \frac{Gm_1m_2}{a^3}\vec{r}_2 = m_1\omega^2\vec{r}_{CM} \quad (1)$$

Here we have assumed origin O as a point fixed in space whose instantaneous position is coinciding with m_1.

Also $$\vec{r}_{CM} = \frac{m_1 \times 0 + m_2\vec{r}_2 + m_3\vec{r}_3}{(m_1+m_2+m_3)} = \frac{m_2\vec{r}_2 + m_3\vec{r}_3}{(m_1+m_2+m_3)} \quad (2)$$

From (1): $$\frac{Gm_1}{a^3}\left[m_2\vec{r}_2 + m_3\vec{r}_3\right] = m_1\omega^2\vec{r}_{CM} \quad (3)$$

Using (2) in (3):

$$\frac{Gm_1}{a^3}\left[(m_1+m_2+m_3)\vec{r}_{CM}\right] = m_1\omega^2\vec{r}_{CM}$$

$$\Rightarrow \boxed{\omega = \sqrt{\frac{G(m_1+m_2+m_3)}{a^3}}}$$

→ Note that if $m_3 << m_2$ and m_1, then this reduces to our familiar binary star formula:

$$\omega = \sqrt{\frac{GM}{a^3}} \quad \text{where } M = m_1+m_2.$$

→ For a binary system, such as Sun Earth system, (O and B say in above diagram), C is a very special point, any satellite located at point C can be at rest in a frame rotating about common CM of m_1 and m_2 and is known as Lagrange point L_4 or L_5. More on that in coming sections.

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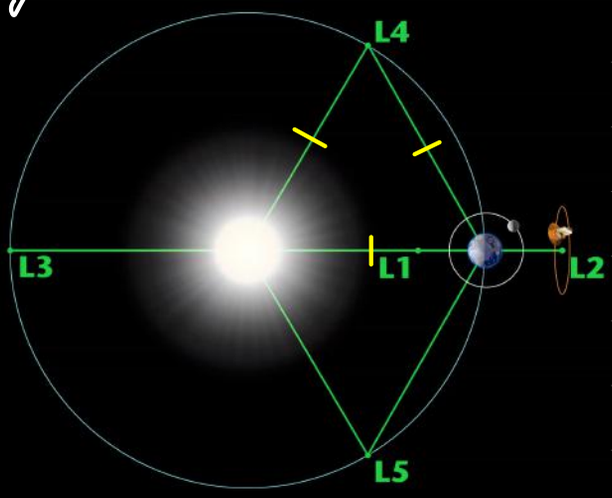
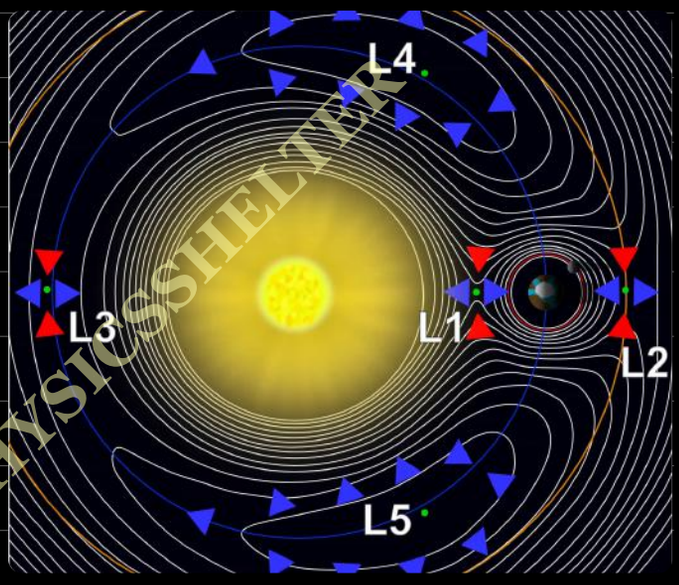
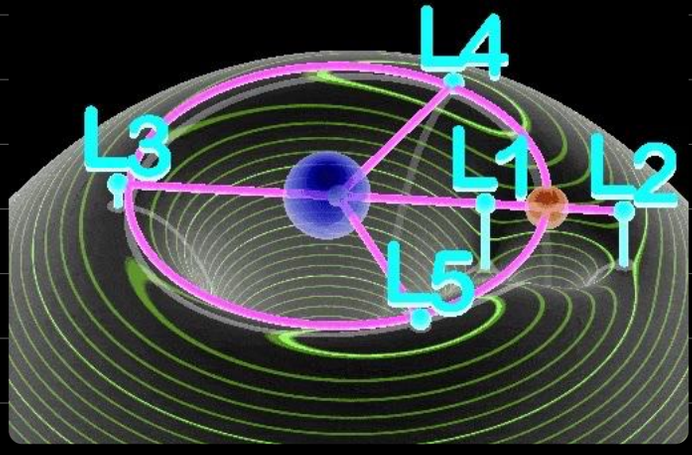
LAGRANGE POINTS: In a two body system rotating in a circle about common centre of mass, the set of points where the resultant of Gravitational forces and centrifugal forces is zero are known as Lagrange points. There are five such points out of which three were discovered by Euler (L_1, L_2 and L_3) and are also known as Euler Points. Out of these points L_1, L_2 and L_3 are points of unstable equilibrium, L_4 and L_5 may be points of stable in mass ratio of sun and planet is about 25 or larger. L_4 and L_5 are located at the vertex of an equilateral triangle whose base is the line segment joining centre of sun and centre of planet. L_4 lies 60° ahead of this line and L_5 lies 60° behind this line. L_4 and L_5 act as junkyards for space debris. L_4 and L_5 points for sun jupiter system has caught some asteroids (Trojans) and they are in stable equilibrium with respect to rotating frame of line joining jupiter to sun.





The famous James Webb Telescope is placed at L_2 point of the earth sun system. It is shielded from sun by the shadow of earth. L_1 offers uninterrupted view of sun and is home to Solar and Heliospheric Observatory Satellite (SOHO). L_3 is not very useful being hidden behind the sun (Except for science fiction stories fantasizing a hidden planet behind the sun. Who knows!?). L_2 also houses Gaia Observatory.



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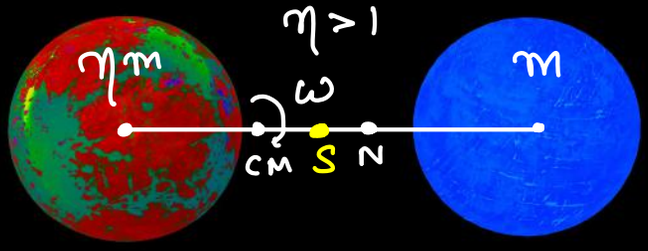
Example : Build-8: Pathfinder

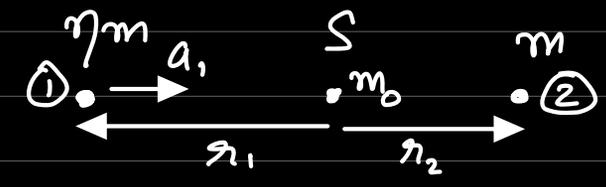
A space ship sent to study a binary star system is set into an orbit in order to remain always collinear with the stars. If distances of the spaceship from the stars are r_1 and r_2, find ratio of masses of the stars. Is this orbit a stable one?



Solution: Based on our discussion from previous section we immediately recognize that the spaceship must be at Lagrange point L_1 and we know beforehand that it must be a point of unstable equilibrium (though we still need to prove it. Now for our system, centre of mass (CM) must be nearer to the heavier star while zero gravity point (N) must be closer to lighter star and our spaceship must be between the CM and N so that centrifugal forces may be able to balance the gravitational force. [To the right of CM, centrifugal force is to the right and to the left of N gravity of heavier star is larger). Now if the spaceship is shifted a little bit to the right from its equilibrium point S, centrifugal force to the right increases and gravitational force to the left decreases thus there is net force to the right. Thus instead of restoring force, the situation is that of unstable equilibrium.

Now to find ratio of masses we shall consider FBD of the spaceship from translating frame of heavier star (ηm) after applying appropriate pseudo forces.

$$a_1 = \frac{G\eta m^2}{(r_1+r_2)^2} \times \frac{1}{\eta m} = \frac{Gm}{(r_1+r_2)^2}$$



$$\omega = \sqrt{\frac{G_1(1+\eta)m}{(r_1+r_2)^3}} \quad \left(\because \omega = \sqrt{\frac{GM}{a^3}}\right)$$ with $M = m_1+m_2$ as per standard binary star theory with circular orbits).

Now drawing the FBD of satellite from frame of m_1:

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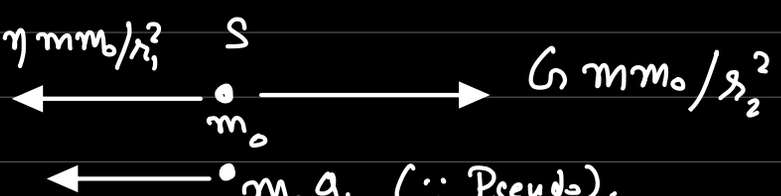


$$\frac{G\eta m m_0}{r_1^2} - \frac{G m m_0}{r_2^2} + \frac{m_0 \cdot Gm}{(r_1+r_2)^2} = m_0 \times \underbrace{\frac{G(1+\eta)m}{(r_1+r_2)^3}}_{\omega^2} r_1$$

Cancelling Gmm_0 throughout:

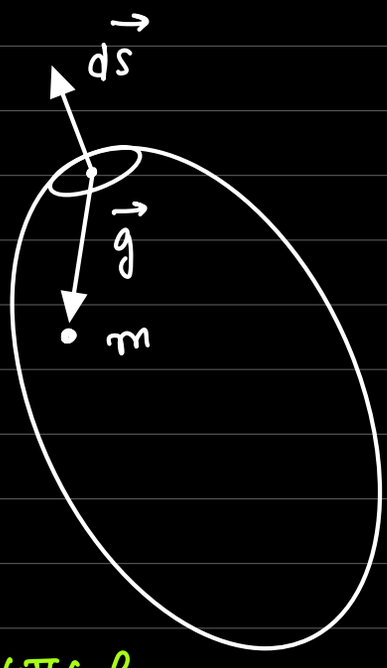
$$\frac{\eta}{r_1^2} - \frac{1}{r_2^2} + \frac{1}{(r_1+r_2)^2} = \frac{(1+\eta) r_1}{(r_1+r_2)^3}$$

Solving for η:

$$\eta = \frac{r_1^3}{r_2^3}\left[\frac{r_1^2+3r_1r_2+3r_2^2}{r_2^2+3r_1r_2+3r_1^2}\right] = \frac{r_1^2}{r_2^2}\left[\frac{(r_1+r_2)^3 - r_2^3}{(r_1+r_2)^3 - r_1^3}\right]$$ Ans.

GAUSS LAW FOR GRAVITATION:

$$\oint \vec{g} \cdot d\vec{S} = -4\pi G m_{in}$$

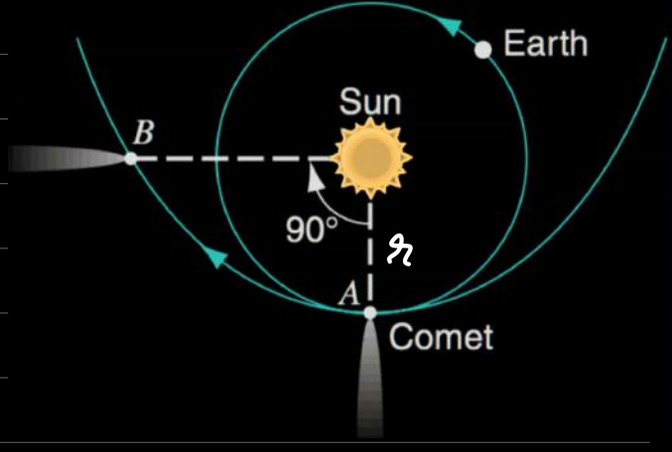


Here $\vec{g}$ is the gravitational field associated with a patch with area vector $d\vec{S}$ and m_{in} is the mass enclosed in the Gaussian surface (An imaginary closed surface for applying Gauss' law).

→ Other Electrostatics Analogues: $\vec{\nabla} \times \vec{g} = 0$, $\vec{\nabla} \cdot \vec{g} = -4\pi G \rho_m$

EXAMPLE: (KEVIN ZHOU):

A comet passes by Sun as shown in a parabolic path. How long in years does comet take in getting from point A to point B.



Solution: Orbital velocity of earth = $\sqrt{\frac{GM}{r}}$

$\Rightarrow$ Velocity of Comet = $\sqrt{\frac{2GM}{r}}$

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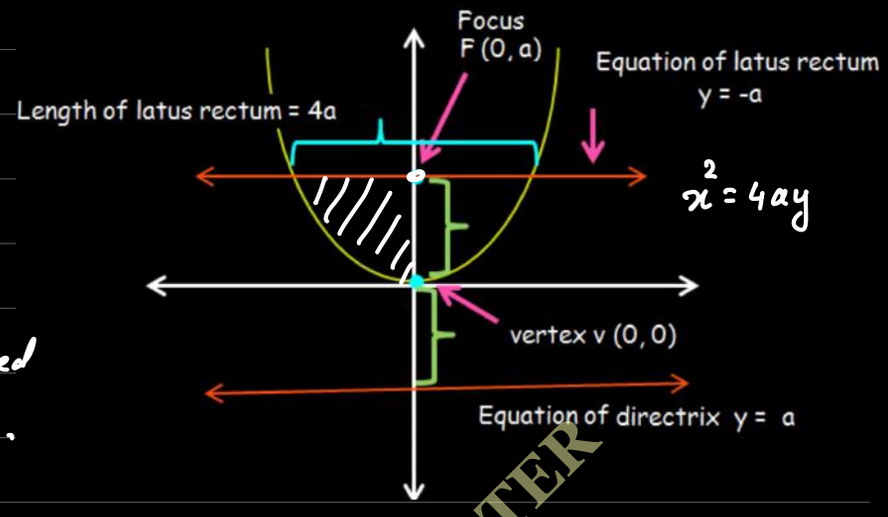
(∵ Parabolic orbit implies escape velocity).

Thus orbital velocity of comet $= L/2m = mR\sqrt{\frac{2GM}{R}} \cdot \frac{1}{2m} = \sqrt{\frac{GMR}{2}}$

Now recall the geometry of a parabola of the form $x^2 = 4ay$

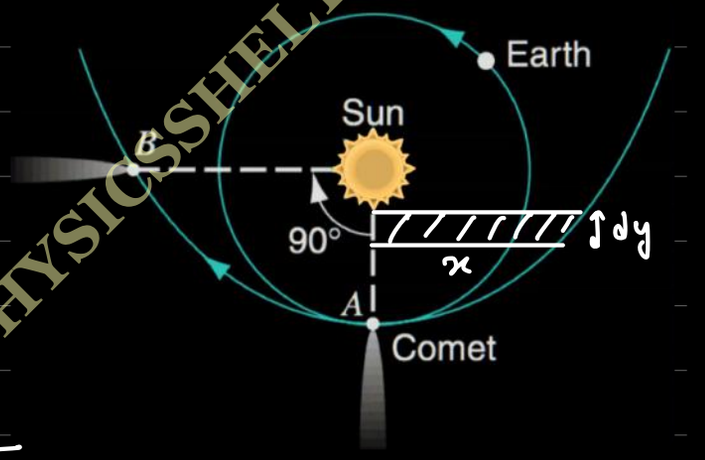
For our problem, $a = R$, and we need to estimate the shaded area. Our parabola is $x^2 = 4Ry$. Consider the shaded strip in the parabolic orbit.

$dA = x\,dy = \sqrt{4Ry}\,dy$



Thus required shaded area:

$$\int_0^R \sqrt{4Ry}\,dy = \frac{4R^2}{3}$$



Thus required time: $A/(dA/dt)$

$$T_c = \frac{4R^2}{3} \times \sqrt{\frac{2}{GMR}}$$

Time of orbit of earth $T_E = \left[\frac{4\pi^2}{GM} R^3\right]^{1/2} \Rightarrow \frac{T_c}{T_E} = \frac{2\sqrt{2}}{3\pi}$

Thus required time: $\frac{2\sqrt{2}}{3\pi}$ year Ans.

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